

Elsewhere in North America: How U.S. Tariffs on China Boosted Mexico's Manufacturing Employment and Output*

Hâle Utar

Grinnell College, IZA, CESifo, RFBerlin

April 25, 2026

Using longitudinal firm- and plant-level data from Mexico linked to customs records over 2014–2023, I examine whether US tariffs targeting China stimulated manufacturing expansion in Mexico. Leveraging the abrupt shift in US trade policy as a natural experiment, and constructing firm-level trade policy exposure measures based on firms' pre-shock trade portfolios, I show that higher US tariffs on China significantly increased manufacturing output, value added, and employment. Foreign multinationals and their domestic affiliates operating under Mexico's export platform, IMMEX, drive these gains, with U.S.-headquartered firms making a particularly notable contribution. Adjustment occurs along both intensive and extensive margins, through expansion of existing plants and establishment of new ones. Employment gains are concentrated in technology-intensive industries and among production workers and technicians—precisely the middle-skill manufacturing jobs that trade protectionism sought to restore domestically. These findings show that heightened import protection stimulated manufacturing activity elsewhere in North America, driven by multinationals, including U.S.-headquartered firms, highlighting how the realized outcomes of trade policy can diverge from their intended effects in an integrated global economy.

Keywords: Trade War, Global Value Chains, Multinational Firms, Manufacturing, Output, Employment, Nearshoring

JEL classifications: F13; F14; F23; F61; F68

*The analysis in this paper was conducted at the INEGI data lab in Mexico City. I am especially grateful to Dr. Natalia Volkow, head of the data lab, for her exceptional support; Gerardo Durand, Deputy Director General of Economic Administrative Records; Lázaro Trujillo Hernández, Director of Economic Statistics of Administrative Records; and the entire INEGI team responsible for the IMMEX and PEME databases, including José Francisco Cuiriz Moncada and others, for many detailed conversations, their ongoing support, and their patience with my limited Spanish. I thank Justin Pierce, Richard Condor for insightful discussions, Stephen Redding, Felix Tintelnot, Kei-Mu Yi and seminar and conference participants at the NBER ITI Spring Meeting, TIGN Santiago, CESifo Global Economy, InsTED (Alberta), Tariffs and Trade Policy: Past, Present, and Future (NC State), Roberts Fest (Penn State), ASSA 2026, MWIEDC Wisconsin-Madison, University of Houston, and UNLV for helpful comments.

1 Introduction

The imposition of U.S. tariffs on a broad range of Chinese imports under Section 301 of the Trade Act of 1974 marked a sharp departure from decades of liberal trade policy. By 2019, the tariffs and subsequent retaliation had affected more than \$450 billion in bilateral trade flows, creating a large and sudden shock to globally integrated production networks (Fajgelbaum and Khandelwal 2022).¹

Although the tariffs were promoted by the administration as a tool to bring back manufacturing jobs (The White House 2018, 2019), recent evidence suggests that they failed to deliver on their central promise of reviving domestic manufacturing (Flaaen and Pierce 2024; Autor et al. 2024), while imposing substantial costs on U.S. consumers and firms (Amiti et al. 2019; Grossman et al. 2024). Instead, studies document sharp increases in multinational-driven exports from Mexico and other third countries, pointing to a broader reorganization of global value chains rather than reshoring (Utar et al. 2025; Fajgelbaum et al. 2024; Beason et al. 2025). Because manufacturing in the United States and Mexico is tightly integrated within North American production networks (Flaaen et al. 2025), the surge in exports from Mexico raises a central question about the mechanism through which U.S. trade policy operates: did the U.S. tariffs on China lead multinational firms to expand manufacturing capacity elsewhere within North America, and if so, where within their cross-border production networks did this adjustment occur?

To examine this question, I use firm-level data from Mexico’s national statistics agency (INEGI) covering internationally active manufacturing firms from 2014 to 2023. INEGI’s Perfil de Empresas Manufactureras de Exportación (PEME) framework links administrative trade records with firm-level information on production and employment, and can be further linked to establishment-level records for firms operating under Mexico’s export platform. The resulting database covers the universe of manufacturing firms active in international trade, which together account for more

¹Studies show that the 2018–2019 tariff changes were uncorrelated with prior price or import trends across products, making the event a suitable natural experiment for analyzing trade policy and global value chains (e.g., Cavallo et al. 2021; Fajgelbaum et al. 2020; Amiti et al. 2020). For details on product and sectoral coverage, see Bown (2021).

than 84% of Mexico’s merchandise trade.

Distinguishing firms that operate under Mexico’s export platform—the Maquiladora Manufacturing Industry and Export Services (IMMEX) program—reveals that IMMEX firms account for 90% of Mexico’s manufacturing exports to the United States. As shown in Utar et al. (2025), these firms, most of which are either subsidiaries of foreign multinationals or domestically owned affiliates closely tied to U.S. manufacturing, are the primary drivers of Mexico’s export growth following the increase in U.S. tariffs on Chinese imports. Throughout the paper, I refer to these firms as “GVC firms,” reflecting the program’s explicit purpose of fostering participation in global production and service networks.

Using corporate linkages from Dun & Bradstreet, I trace the ultimate owners of GVC firms via their tax identifiers and link this information to the PEME database. This allows a direct assessment of the role of foreign ownership in Mexico’s manufacturing expansion. Between 2017 and 2023, firms with ultimate owners headquartered outside Mexico created over half a million manufacturing jobs, of which U.S.-owned firms alone expanded employment by 31 percent, adding more than 300,000 jobs (Figure 1). This aggregate pattern is echoed in a number of high-profile multinational expansions—including Bosch, LEGO, Continental, Volkswagen, and Toyota— that committed substantial investments to expand their manufacturing footprint in Mexico during this period (News Desk 2018; Webwire 2022; Reuters 2022a; Cohen et al. 2025). In several cases, these expansions coincided with reductions in US operations, as seen with John Deere, Foxconn, and Stellantis (Reuters 2020; Reuters 2022b; Glickman 2024).

Did a policy aimed at reviving US manufacturing instead fuel job creation in Mexico? In fact, the share of employment accounted for by manufacturing firms in IMMEX relative to US manufacturing employment rose from roughly one-fifth in the pre-trade war period to one-fourth in the post-trade war period.²

² Author’s calculation based on employment of IMMEX firms in the PEME database relative to U.S. manufacturing employment from the Federal Reserve Economic Data (FRED), Federal Reserve Bank of St. Louis. Pre-trade war period covers 2014–2017; post-trade war period covers 2018–2023. Manufacturing employment accounted for by IMMEX firms in PEME grew from 2.2 million in 2014 to 3.3 million in 2023.

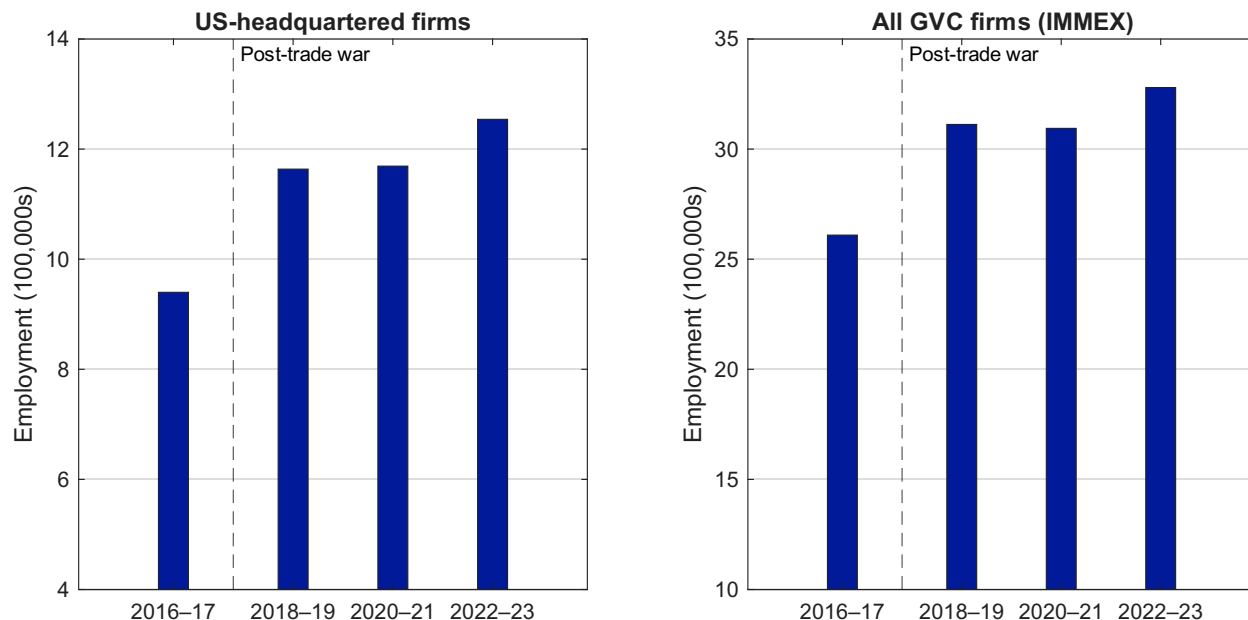


Figure 1: Manufacturing Employment accounted for by U.S. Firms in Mexico

PEME database, INEGI. Within PEME, IMMEX firms and US-owned IMMEX firms are identified via their tax identifiers. The figure plots the total employment accounted for by firms whose ultimate owners are headquartered in the US (left) and the total employment accounted for by all IMMEX firms.

I construct firm-level trade policy exposure measures based on pre-shock product-destination portfolios at the six-digit HS level. Exploiting the abrupt and large-scale nature of the US tariff shock, I analyze within-firm outcomes in an event study framework, controlling for aggregate, industry, and firm-specific shocks.

I show that the U.S. trade policy shift against China not only boosted Mexican firms' exports to the United States but also generated significant increases in domestic production and employment, with subsidiaries and affiliates of firms headquartered in the United States and other major economies driving much of this manufacturing expansion.

Distinguishing between contract manufacturing for foreign clients and in-house production, I show that the observed production expansion reflects the relocation of production within firms' cross-border networks, rather than growth in contract manufacturing (*maquila*). This distinction matters because contract manufacturers, by concentrating on the labor-intensive tasks of U.S. firms' production processes, move in tandem with rather than substituting for U.S. manufacturing (Bergin et al. 2009), whereas in-house expansion by multinationals can substitute for it directly. Consis-

tent with production deepening rather than simple trade pass-through, value added rises alongside output, with domestic contributions arising primarily through labor rather than upstream input production.

The employment gains are concentrated in technology-intensive industries — computers, electronics, machinery, aerospace, and automotive manufacturing — where foreign multinationals in North American supply chains are disproportionately active, while food and textile manufacturing see near-zero effects. Within plants, tariff-induced employment growth is concentrated among production workers and technicians, while non-production employment remains largely unaffected. These are precisely the middle-skill manufacturing jobs that trade protectionism sought to restore domestically.

Adjustment occurs along both intensive and extensive margins. Beyond expansion at existing plants, U.S. and non-U.S. foreign firms embedded in North American value chains established new manufacturing plants in response to U.S. tariff hikes targeting China. The effect on new plant entry is 2.5 times larger among foreign-owned firms than domestic firms, and is strongest among U.S.-headquartered firms operating in industries that both compete with Chinese goods in the U.S. market and depend on Chinese inputs, suggesting that North American manufacturing, as a whole, responded in ways that may have partially offset the intended reshoring of production to the United States.

Recent studies highlight that U.S. dependence on Chinese supply chains often persisted through third-country trade flows, raising questions about whether tariff-induced trade diversion translated into domestic production gains (Alfaro and Chor 2023; Freund et al. 2024; Garred and Yuan 2025; Gopinath et al. 2025). This paper provides direct causal evidence that the 2018–19 U.S. tariffs on China significantly increased manufacturing output and employment in Mexico and, to my knowledge, is the first to show at the firm level that these tariffs boosted production within the North American manufacturing network.³ By linking firm-level production data with corporate

³Focusing on firms' trade, the positive effect of the US-China trade war on third-country firms is also shown by Rotunno et al. (2023) and Chen et al. (2025), among others. For the reverse effect, how improved Chinese access to the U.S. market negatively affected Mexican firms, see Utar and Ruiz (2013).

ownership information and documenting capacity expansion along both intensive and extensive margins, the paper addresses key dimensions—which firms drive relocation and how they adjust capacity—that Fajgelbaum and Khandelwal (2022) identify as largely absent from the trade-war literature.

Although U.S. policymakers aimed to revitalize domestic manufacturing through the trade policy shift against China, recent research finds no measurable impact on U.S. output or employment (Flaaen and Pierce 2024; Autor et al. 2024). The findings of this paper help explain this puzzle: the adjustment occurred primarily outside the United States, as multinational firms reorganized production within their cross-border networks. Strikingly, U.S.-headquartered firms themselves were at the center of this reallocation — the very firms the policy sought to incentivize to expand domestically. This emphasizes how the realized outcomes of trade policy can diverge from their intended effects in an integrated global economy.

A central reason for this divergence is that trade is highly concentrated in the hands of large multinational enterprises. Handley et al. (2025) show that the 2018–19 tariffs disproportionately fell on the largest firms accounting for more than 80% of U.S. exports. Flaaen et al. (2025) emphasize that production in the United States is deeply embedded in global value chains, with extensive cross-border back-and-forth linkages with Mexico and Canada. These interdependencies imply that domestic effects of tariffs can differ substantially from conventional predictions: domestic content of imports and foreign content of exports may limit or even reverse the potential gains for domestic producers. Moreover, such back-and-forth trade often takes place within firms, giving multinational enterprises the flexibility to move production across borders in response to policy changes (Conconi et al. 2025; Graziano et al. 2024; McCaig et al. 2025). Consistent with this, Flaaen et al. (2020) highlight tariff-jumping strategies among multinational firms to circumvent import restrictions, and Utar et al. (2025) show that firms in Mexico embedded in global value chains increased their two-way trade with firms in the United States in response to the U.S.–China trade war. Building on this literature, this paper contributes new evidence by showing that foreign-headquartered firms — especially U.S.-owned firms — were at the center of the U.S. trade-policy-

induced manufacturing growth in Mexico. With the expansion driven primarily by in-house rather than contract manufacturing, the reallocation reflects decisions made within large multinationals' cross-border networks⁴, further highlighting how multinational firms' cross-country production networks shape the way trade policy shocks are transmitted across borders.

By showing that U.S. tariffs disproportionately increased factory-floor employment in technology-intensive sectors in Mexico — precisely the middle-skill manufacturing jobs that trade protectionism sought to restore domestically — the findings provide firm-level evidence on the production and employment reallocations underlying the documented welfare losses from the 2018–19 U.S. trade policy (Amiti et al. 2019; Fajgelbaum et al. 2020; Grossman et al. 2024). These findings also speak to recent work on optimal trade policy design in the presence of global supply chains and multinational production, where the welfare implications of tariffs depend critically on how firms reorganize across borders (Antràs et al. 2022; Grossman et al. 2024; Gutiérrez et al. 2025; Blanchard et al. 2026).

The next section introduces the institutional context and describes the firm- and plant-level data. Section 3 presents the tariff exposure measures and empirical strategy. Section 4 presents firm- and plant-level evidence on the effects of U.S. tariffs on Mexican manufacturing exports, output, employment, and capacity expansion, with a particular focus on the role of U.S.-headquartered multinationals. Section 5 concludes. Supplementary results appear in the online appendix.

2 Institutional Background and Data

Deep integration with the United States, first under NAFTA and now under the United States–Mexico–Canada Agreement (USMCA), has made Mexico a central platform for U.S.-centered manufacturing production networks. Competitive labor costs and geographic proximity with a long shared border have supported dense cross-border supply chains in which intermediate in-

⁴Consistent with theories of global sourcing (Antràs and Helpman 2004), the two modes sort firms by productivity and scale, with larger, more productive multinationals choosing vertical integration over arm's-length contracting, suggesting the reallocation is concentrated among larger firms rather than reflecting a broad-based response across the U.S. firm size distribution.

puts cross the border multiple times before final assembly. Recent work using confidential U.S. establishment-level data shows that such global value chain linkages are both large and increasingly important and highlight that U.S. manufacturing performance is closely tied to back-and-forth trade with nearby partners, especially Mexico (Flaaen et al. 2025). Yet much of the empirical literature identifies firms embedded in GVCs simply as those that both import and export (Antràs and Chor, 2022), without reference to the institutional arrangements that organize this production sharing. In Mexico, the key institutional framework is the IMMEX program, which allows firms to temporarily import intermediate inputs duty-free for processing and re-export.

2.1 The Creation of Mexico's Export Platform–IMMEX

In 2007, following the full implementation of NAFTA, which eliminated duty-free access to non-NAFTA inputs among export-processing establishments, Mexico's Ministry of Economy consolidated all export promotion schemes under a single framework known as the Maquiladora Manufacturing Industry and Export Services Program (IMMEX).

The program is designed to support firms integrated into regional and global value chains by lowering transaction costs on inputs not destined for domestic consumption, and has become the principal vehicle through which foreign firms enter and run manufacturing operations in Mexico for the broader North American market. IMMEX establishments are predominantly multinational subsidiaries active in North American supply chains, alongside contract manufacturers serving U.S. firms.

Firms register under IMMEX to access a set of trade-facilitating benefits, including a waiver of the 16 percent value-added tax typically levied on all purchases, including imported inputs and capital equipment, reduced custom processing fees, and the option to defer tariff payments on inputs not originating in North America until the time of re-export. To qualify, firms must either maintain an annual export value of at least 500,000 USD or derive at least 10 percent of their revenue from exports. In 2017, firms operating under IMMEX accounted for 84 percent of Mexico's total exports and 64 percent of its imports and provided 2.7 million jobs.

2.2 Data

Manufacturing Firms Matched to Customs Transactions

The PEME database links transaction-level customs records detailing firms' export and import activities with firm-level information on manufacturing production and employment. PEME covers the universe of exporting manufacturers observable in INEGI's manufacturing statistical system, with no size threshold, spanning from large multinationals to small and micro firms.

Production and employment information at the firm level are aggregated from establishment-level data considering only establishments whose primary activity falls within manufacturing. Each customs record includes product codes classified under Mexico's tariff schedule (TIGIE), which closely aligns with the Harmonized System at the six-digit level, destination or origin countries, and transaction values reported in USD. Firms' customs data are linked with the six-digit products and country pairs subject to newly imposed tariffs as part of the 2018–19 trade war, using the datasets of tariff changes on US imports and exports assembled and updated by Fajgelbaum et al. (2020).⁵

In addition to its near-universe coverage of exporting manufacturing firms, the PEME provides comprehensive coverage of workers, including both directly hired employees and workers supplied through third-party staffing firms. This is particularly important in the Mexican manufacturing sector, where staffing arrangements are widespread and lead social security–based employment records to systematically understate employment and labor compensation. Estefan et al. (2024) show that such arrangements account for nearly one-fifth of manufacturing employment and generate substantial gaps in worker coverage and wage reporting in data from the Mexican Social Security Institute (IMSS).⁶

⁵To calculate the tariff changes in HS 6-digit goods, I use the share of each HS 10-digit good per source country in total US HS 6-digit import in the pre-trade war year, 2017, as a weight to collapse the US import tariffs data into the HS6 level. Similarly, I calculate the share of each HS 8-digit good per country in total US HS 6-digit export as a weight to collapse the retaliatory tariff data into the HS6 level. See Appendix B.

⁶Manufacturers frequently hire workers through third-party staffing firms to reduce payroll taxes, profit-sharing obligations, and exposure to severance and unionization costs, registering them at the statutory minimum wage with the remainder of compensation paid through non-contributory bonuses.

The 2017 PEME dataset includes 6,092 exporting manufacturing firms, which together accounted for approximately 3.5 million jobs. Merging PEME with the Ministry of Economy’s IMMEX registry identifies 4,092 firms operating under the IMMEX export program, representing about two-thirds of the firms and accounting for 76% of employment in the export manufacturing sector.

Table 1: Summary Statistics—Estimation Sample

Variable	(1) Mean	(2) SD	(3) p25	(4) p75	(5) Obs
Panel. A					
All 2017 Exporting Firms (PEME)					
Log Employment	5.44	1.46	4.51	6.39	52,198
Log Value of Production	16.65	1.86	15.49	17.92	51,865
Log Value of Exports	15.09	2.82	13.38	17.08	50,616
Log Value of Imports	15.27	2.59	13.69	17.08	49,806
US Export Share	0.75	0.36	0.60	1.00	50,616
Number of Plants	1.55	3.85	1.00	1.00	51,679
Panel. B					
	IMMEX		Others		
	Mean	Obs	Mean	Obs	
Log Employment	5.47	37,391	5.38	14,807	
Log Value of Production	16.62	37,209	16.73	14,656	
Log Value of Exports	15.62	36,872	13.67	13,744	
Log Value of Imports	15.53	36,465	14.56	13,341	
Number of Plants	1.46	36,874	1.77	14,805	

Notes: Manufacturing exporters as of 2017 over 2014-2023. Values are expressed in 2019 constant USD. Employment is measured in number of workers; number of plants is reported in levels.

Table 1 summarizes the estimation sample, which consists of exporting firms in 2017. Panel B reports firm characteristics separately for GVC (IMMEX) firms, which are identified by Utar et al. (2025) as the primary drivers of Mexico’s export response to the U.S.–China trade war, and other exporting manufacturing firms. The median firm in the PEME data employs 239 workers, while

the median GVC and non-GVC firms employ 245 and 228 workers, respectively. Accordingly, average log employment is very similar across the two groups—5.5 among GVC firms and 5.4 among non-IMMEX exporters. Although employment is slightly higher in GVC firms, non-GVC exporters exhibit larger production volumes, whereas GVC firms export more, consistent with GVC firms being more labor-intensive and export-focused than their non-GVC counterparts.

Figure A.1 shows the distribution of firms across two-digit manufacturing industries, separately for GVC firms and for exporting manufacturing firms outside IMMEX. GVC firms are concentrated in the machinery and equipment industry, reflecting North American supply chain specialization in capital-intensive manufacturing. Using firms' unique tax identifiers, I link firms operating under IMMEX in the PEME data to D&B's Hierarchy and Connections database to identify each firm's global ultimate parent and its country of headquarters. Slightly more than half of the GVC firms are foreign-owned, and these foreign firms employed roughly 2 million workers in 2017.⁷ Figure A.2 shows that foreign ownership is strongly concentrated in machinery and equipment, particularly among U.S.-owned firms. Table A.2 presents summary statistics separately by ownership status, revealing that U.S.-owned firms are the largest employers while non-U.S. foreign firms lead in production volume, with domestic IMMEX firms smaller across both dimensions.

To complement the firm-level PEME data, I also draw on INEGI's plant-level IMMEX survey of manufacturing establishments covering 2015–2023, which provides the detail needed to distinguish in-house production from contract manufacturing, compute value added, and examine within-plant employment adjustment. The survey reports comprehensive measures of revenues, input costs, and labor inputs at the establishment level, with sales of own manufactured goods and contract manufacturing revenues reported separately and disaggregated by destination (domestic versus foreign).

Table A.3 reports summary statistics for IMMEX establishments matched to firms in PEME. The average labor cost share is 21%, while material costs account for 62% of total expenses. A majority

⁷Firms whose ultimate owner is not identified as foreign also include *shelter firms*—firms operating under contractual arrangements with foreign clients—as well as firms affiliated with foreign companies but without a direct ownership link, or those owned by a foreign company holding less than 50 percent of shares.

of material inputs originate outside Mexico, as reflected in a foreign material cost share of 42%.

3 Empirical Strategy

To identify the causal impact of the U.S.–China trade war on manufacturing firms in Mexico, I construct firm-level tariff exposure measures from firms’ six-digit product level exports and imports in the pre-shock year 2017. I then estimate how firms’ outcomes evolved differentially across years according to their susceptibility to the trade shock.

3.1 Firm-Level Exposures to the US-China Trade War

3.1.1 US Import Tariffs on China

The analysis centers on the 2018–19 US import tariffs imposed on China, which covered approximately \$350 billion of U.S. imports. Let $USIT^{CN}$ represent the set of HS 6-digit products subject to the 2018–19 US import tariffs on China and let $\Delta\tau_\ell^{USIT^{CN}}$ denote the change in the US import tariff rate for HS 6-digit good ℓ from China, measured in percentage points and expressed as a decimal.⁸ Additionally, let $X_{i\ell}^{2017}$ be the value of exports of firm i in good ℓ in year 2017. Then, the level of exposure of firm i in Mexico to the increased US import tariffs on Chinese goods can be determined as:

$$T_i^{US \rightarrow CN} = \frac{\sum_{\ell \in USIT^{CN}} X_{i\ell}^{2017} \times \Delta\tau_\ell^{USIT^{CN}}}{\sum_{\ell} X_{i\ell}^{2017}} \quad (1)$$

Here, $X_{i\ell}^{2017}$ denotes firm i ’s exports of good ℓ in 2017, and $T_i^{US \rightarrow CN}$ measures the tariff-weighted share of targeted goods in the firm’s overall exports.

While most exports from a typical Mexican firm are destined for the United States (see Table 1), Equation 1 does not distinguish export destinations. This is because the increased cost of Chinese

⁸Since the tariff escalations occurred in multiple phases between 2018 and 2019, I use the increased tariff rates valid as of December 2019.

goods entering the US market is expected to make the US more attractive to Mexican exporters relative to other markets. As a result, manufacturing firms that previously produced and exported affected goods to non-US markets are potentially treated by the increased appeal of the US market. Consider two manufacturing exporters with identical export product portfolios and the same level of $T^{\text{US} \rightarrow \text{CN}}$, say 0.20. Suppose one sells 50 % of its exports to the US while the other sells 100 %. The firm exporting entirely to the US is treated via its full customer base, whereas the other firm is treated via only half of its customer base. At the same time, the firm selling 50% of its export capacity to the US may have greater flexibility to expand its US exports without going through costly adjustment of its production capacity. Thus, when focusing on firms' trade outcomes, it is important to adopt a broader definition of treatment.

However, firms with an established customer base in the US are more likely to respond to the costlier entry of Chinese goods to the US by expanding their production capacity. To account for this in the analysis of firms' output and employment outcomes which is the focus of this paper, exposure to US tariffs targeting China is constructed specific for the US market, as follows:

$$T_{i,\text{US}}^{\text{US} \rightarrow \text{CN}} = \frac{\sum_{\ell \in \text{USIT}^{\text{CN}}} X_{i\ell}^{2017,\text{US}} \times \Delta \tau_{\ell}^{\text{USIT}^{\text{CN}}}}{\sum_{\ell} X_{i\ell}^{2017}} \quad (2)$$

where $X_{i\ell}^{2017,\text{US}}$ denotes firm i 's US exports of good ℓ in 2017.⁹

The exposure measure in Equation 2 is referred to as *narrow* since it classifies firms with no pre-shock US exports as unexposed, even if their export portfolios are concentrated in targeted goods. However, firms with high broad exposure but little or no pre-shock US exports may have redirected exports to the US market following the tariff shock, which would bias the estimated effect downward. Appendix Table B.1 confirms this concern: firms with initially low US export shares increased their US export share more than others following the shock. To address this, when Equation 2 is used as the exposure measure, I restrict the sample to firms that were exporting to

⁹One can also use the measure described in Equation 1 and scale it by the firm's pre-shock U.S. export share, thereby assigning greater weight to U.S.-oriented firms. This alternative approach yields similar results.

the US in 2017, defined as those deriving at least 30% of export revenue from the US market.¹⁰

To account for confounding effects of firms' exposure to concurrent trade policy changes, namely the US–Mexico tariff escalation under Section 232, and China's retaliatory tariffs, I construct firm-specific measures that reflect firms' susceptibility to these policies. I further consider the potentially confounding influence of tariffs on firms via the input channel and construct exposure measures based on the relative importance of targeted goods in firms' imports from China and the US. These additional measures are described in Appendix Section B.2

3.2 Empirical Model

Using the firm-level trade policy exposure measures constructed in Section 3.1, I estimate the effects of the U.S. import tariff increases targeting China on manufacturing firms in Mexico. The analysis is conducted on the universe of firms with positive exports in 2017 and at least one employee. I examine firms' trade, production, and employment responses using an event study framework covering the ten year period from 2014 to 2023. The analysis begins in 2014 to establish a counterfactual baseline four years prior to the onset of the trade war and extends through 2023 to capture long-run effects.

Identifying the Impact of the U.S. Tariffs Hikes Targeting China on Firms in Mexico

Let Y_{ijt} be firm i 's outcome variable in industry j at time t , for example, the log value of employment. To examine the effects of the US tariffs targeting China on manufacturing firms in Mexico, the following generalized difference in differences (DD) equation with two sided fixed effects is formed:

$$Y_{ijt} = \beta_0 + \sum_{\substack{h=2014 \\ \neq 2017}}^{2023} \beta_h \mathbb{1}_{h=t} \times T_i^{\text{US} \rightarrow \text{CN}} + P_{it} + \eta_{jt} + \phi_i + \varepsilon_{it} \quad (3)$$

¹⁰The 30% threshold is informed by the results in Table B.1 which reports the average treatment effect on firms' US export shares based on the broad exposure measure, while sequentially excluding firms with low baseline US export shares. The estimated tariff exposure effect on U.S. export share becomes statistically insignificant once the sample is restricted to firms with at least 20% of exports destined for the United States. I adopt a 30% cutoff as a conservative criterion that also preserves a sufficiently large sample.

In Equation 3, the continuous firm-level treatment variable, $T_i^{\text{US} \rightarrow \text{CN}}$, is interacted with year indicators to examine the yearly evolution of the impact. Using 2017 as the baseline year, I track exporting manufacturing firms through 2023 to assess the effects of the US tariff hikes targeting China, while also looking backward until 2014 to identify any pre-existing trends.

The vector P_{it} controls for concurrent trade policy changes that could confound the estimated effect of US tariffs targeting China. In the baseline specification, it includes interactions with year fixed effects of the US-Mexico tariff escalation exposures, namely $T_i^{\text{US} \rightarrow \text{MEX}}$ (Equation B.1), and $T_i^{\text{MEX} \rightarrow \text{US}}$ (Equation B.2). This allows for differential time trends for firms susceptible to US tariffs on Mexican goods and firms whose US imports are affected by Mexico’s retaliatory tariffs on US goods. P_{it} further accounts for firms’ exposure to US tariffs on China through their imported inputs and to China’s retaliatory tariffs on US goods (Equations B.3-B.6).

The equation also includes η_{jt} , which denotes three-digit industry by time fixed effects to control for potentially confounding time trends that may vary across narrow industries. For example, the key change introduced by the USMCA in 2020 was the tightening of local content requirements in the automotive and textile sectors, and these changes are absorbed by the fixed effects.¹¹ As a result, the tariff effect is identified from variation across firms within three-digit industries as industry-wide variations are contained by η_{jt} .¹²

The time-invariant or relatively persistent firm characteristics such as management skills, production technology, and line of business are controlled with firm fixed effects, ϕ_i . This means that coefficients are estimated from within-firm variation over time as the influence of any observed or unobserved initial firm characteristic that may be correlated with firm i ’s future exposure to the change in the US trade policy is absorbed by firm fixed effects. The error term ε_{it} is assumed to have a zero mean, with standard errors clustered at the firm level to allow for within firm correlation, reflecting the level at which the treatment variable varies (Abadie et al., 2023).

¹¹ As a robustness check, Appendix Figure D.4 shows that production and employment effects remain strong—and if anything, marginally stronger—when excluding firms prone to USMCA provisions.

¹² In Appendix Section D, Figures D.1–D.3 illustrate how the estimates change as control variables are added sequentially.

Plant-level Analysis

Complementing the firm-level analysis, I estimate the following plant-level event-study specification using data on IMMEX establishments over the period 2015–2023:

$$Y_{pijt} = \alpha_0 + \sum_{\substack{h=2015 \\ \neq 2017}}^{2023} \alpha_h \mathbb{1}_{h=t} \times \mathbf{T}_i^{\text{US} \rightarrow \text{CN}} + P_{it} + \eta_{jt} + L_{pt} + \phi_p + \varepsilon_{pt} \quad (4)$$

In Equation 4, ϕ_p denotes plant fixed effects and, as before, η_{jt} represents three-digit industry-by-year fixed effects. Here, the industry classification is based on each plant p 's primary activity, rather than firm i 's. Leveraging plant location information available in the plant-level database, L_{pt} includes municipality-by-year fixed effects to account for potentially confounding effects of the minimum wage reform enacted in 2019, which raised the minimum wage in border municipalities by 100 percent.¹³ Since municipalities in Mexico are typically contained within local labor markets (Utar, 2025), controlling for L_{pt} also absorbs local labor market shocks, including those associated with labor shortages and COVID-related shutdowns.

The treatment variable remains defined at the firm level, and the vector P_{it} continues to include controls for concurrent trade policy changes and tariff exposures through firms' imported inputs, as in Equation 3. The error term ε_{pt} is assumed to have mean zero, and standard errors are clustered at the firm level, which is the level at which the treatment variable varies.

To estimate average treatment effects rather than year-by-year dynamic responses, I replace the event-study interactions in Equations 3 and 4 with interactions between the firm-specific trade exposure measures and a post-shock indicator, PostTW_t , equal to 1 from 2018 onward and 0 otherwise.

¹³In 2019, Mexico revised its minimum wage legislation to improve workers' living conditions. The reform was implemented unevenly, dividing the country into two regions: the northern border region—which includes 45 municipalities located within approximately 25 kilometers of the U.S. border—and the rest of the country. In the northern border region, the minimum wage doubled in 2019, while in the rest of the country it rose gradually, with annual increases of 16%, 20%, and 15%, respectively, between 2019 and 2021.

4 Production and Employment Consequences of the US Trade Policy Shift Against China

This section uses firm-level production and employment data linked to customs records through 2023 to confirm the persistent positive effect on firms' exports to the U.S., then address the key question of whether this trade response translated into domestic production and employment gains. I then examine the nature of this expansion: which workers benefited, whether firms adjusted along the extensive margin through new plant openings, and the central role of U.S.-headquartered multinationals in driving these gains.

4.1 The Effect of U.S. Tariffs on China on Mexican Manufacturing Firms' Exports

I begin by estimating Equation 3, where the dependent variable is the logarithm of firms' exports to the United States. Figure 2 presents the year-by-year DD coefficient estimates for the continuous firm-level treatment variable $T_i^{US \rightarrow CN}$.

The results show that higher U.S. tariffs on Chinese imports significantly increase manufacturing firms' exports to the United States. The DD estimates are statistically indistinguishable from zero prior to the onset of the trade war, indicating no meaningful differential pre-trends across firms with differing exposure levels. The estimates then rise gradually relative to the 2017 baseline, reaching 1.71 by 2023, with a temporary drop during the COVID pandemic in 2020.

To gauge the economic magnitude of the effect, note that firms at the 90th percentile of tariff exposure faced a 25% increase in U.S. tariffs on Chinese imports, compared with 7.7% for firms at the 10th percentile (see Table A.1). This difference in exposure implies a 34% increase in exports to the United States ($e^{1.71 \times 0.17}$).¹⁴

¹⁴Appendix Figure C.1 shows that tariff exposure also increased firms' total exports and total imports, indicating an overall expansion in trade activity.

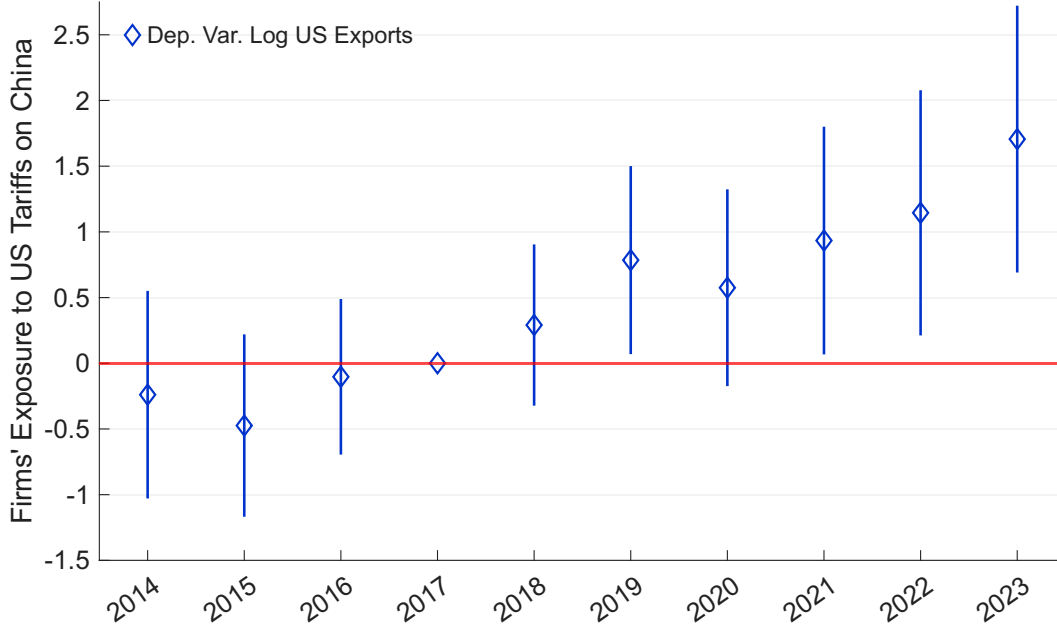


Figure 2: Impact of US Tariffs Targeting China on Manufacturing Firms' Exports

Notes: Estimation of Equation 3. The dependent variable, as indicated in the legend, is the natural logarithm of firms' value of exports. PEME firms exporting as of 2017, covering the period from 2014 to 2023, with $N=46,453$. The regression controls for firm fixed effects, three-digit-industry-by-year fixed effects and concurrent trade policy changes. The y-axis displays the yearly DD coefficient estimates corresponding to $\mathbf{T}_i^{\text{US} \rightarrow \text{CN}}$, along with 95% confidence intervals.

These results confirm and extend Utar et al. (2025), who document a positive effect through 2021 using customs data and attribute it primarily to GVC firms operating under IMMEX. Figure 3 shows that both findings hold through 2023: the U.S. trade policy shift has a significant positive effect on GVC firms, with the estimated coefficient reaching 1.92 by 2023, while the impact on non-IMMEX manufacturing firms remains statistically insignificant throughout 2018–2023. The coefficient estimate of 1.92 corresponds to a 39% increase in U.S. exports for firms at the 90th percentile of exposure relative to those at the 10th percentile.

To address the possibility that the observed export growth reflects transshipment of Chinese goods through Mexico, I construct a measure of firms' U.S. exports net of Chinese imports, defined as

$$NetUSX_{it}^{MCN} = \sum_{j \in J_{it}^{\text{USX}}} [X_{ijt}^{\text{US}} - M_{ijt}^{\text{CN}}],$$

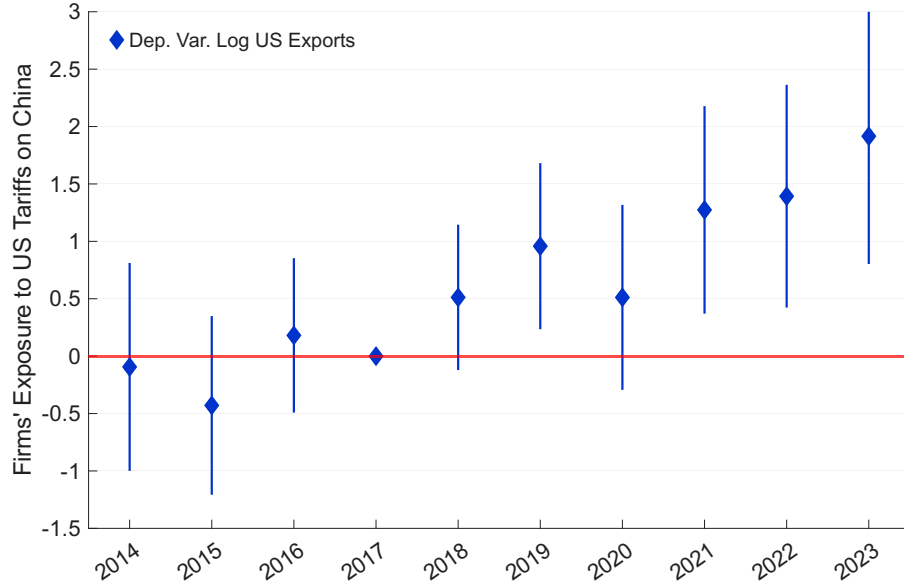
where J_{it}^{USX} denotes the set of six-digit HS products that firm i exports to the United States in

period t . This measure captures the value of firm i 's U.S. exports net of its imports from China of the same products. Appendix Figure C.2 shows that the estimated effect on U.S. exports net of Chinese imports closely follows the effect on gross U.S. exports, indicating that the export response is not driven by simple transshipment of Chinese goods.

Table C.1 in the Appendix reports the average treatment effects on firms' trade outcomes, separately for GVC firms and other manufacturing exporters (firms outside the IMMEX program). The results show that the broader trade response to U.S. tariffs on China is also concentrated among GVC firms. In addition to increasing their exports to the United States, GVC firms significantly expand their imports and total exports, whereas non-IMMEX manufacturing firms show little response. The estimates also show that, on average across GVC firms, the share of imports sourced from China does not increase in response to the tariff hikes.

These findings indicate that the export response to the U.S. trade policy shift is primarily driven by firms operating under Mexico's IMMEX program, which encompasses firms integrated into North American production networks. Importantly, the effect reflects who operates within IMMEX, rather than the program's specific policy features. To examine the role of foreign multinationals in the export response, I distinguish firms by ownership using Dun & Bradstreet's corporate hierarchy data and re-estimate the effects separately among foreign-owned IMMEX firms. With an estimated coefficient of 2 in 2023, the results in Figure C.3 in the Appendix show a strong export response of foreign-owned firms following the U.S. trade policy shift, confirming that foreign multinationals are at the center of Mexico's export expansion.

Taken together, these results show that firms more exposed to U.S. tariffs on Chinese goods significantly increased their exports to the United States, with the export gains from the U.S.–China trade war largely accounted for by firms embedded in global value chains, many of which are foreign multinationals operating in Mexico. Did this expansion in exports and imports translate into higher manufacturing output in Mexico? I now turn to the effects on production.



(a) GVC Firms



(b) Other Manufacturing Firms

Figure 3: Impact of US Tariffs Targeting China on Manufacturing Firms' Exports

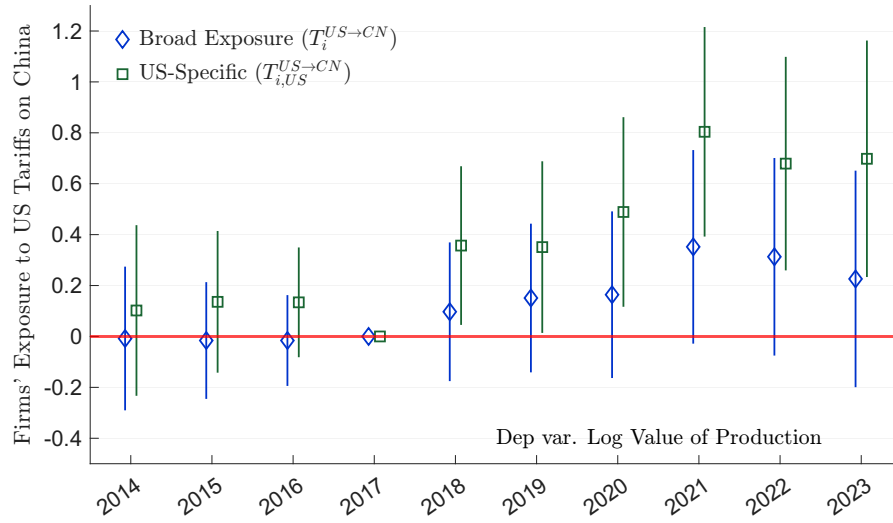
Notes: Estimation of Equation 3. PEME database. N=35,787 (IMMEX firms in PEME) N=10,666 (Others). The dependent variable in both figures is the natural logarithm of firms' value of US exports. All regressions control for firm fixed effects, three-digit-industry-by-year fixed effects and concurrent trade policy changes. The y-axis displays the yearly DD coefficient estimates corresponding to $T_i^{US \rightarrow CN}$, along with 95% confidence intervals.

4.2 How US Tariffs on Chinese Products Stimulated Manufacturing Output in Mexico

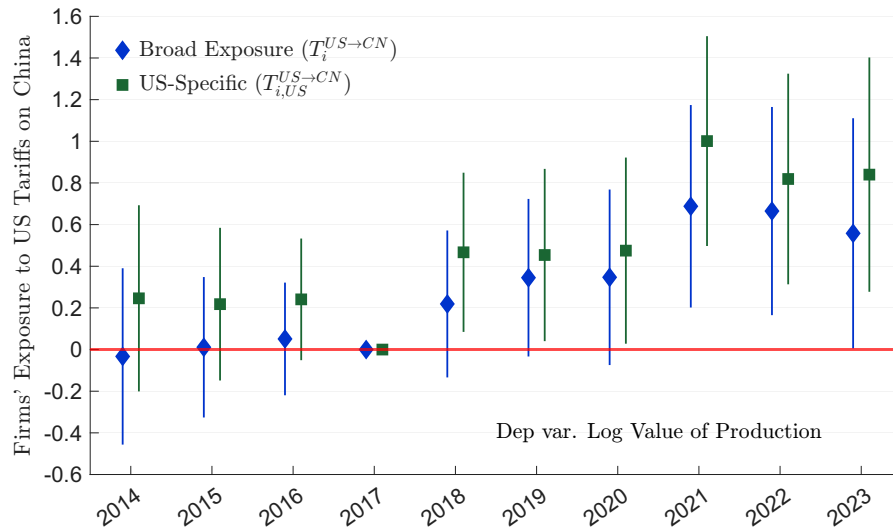
I begin by examining how firms' manufacturing output, measured as the value of production at factory prices, responds to the U.S. trade policy shift. Because expansion may be more pronounced among firms with an established customer base in the United States, I estimate the effect of U.S. tariffs on manufacturing output using both the narrow U.S.-specific measure, estimated among firms exporting to the United States in 2017, and the broad measure, estimated among all firms that exported in 2017. Figure 4 presents these results, with Panel (a) showing estimates for all exporters and Panel (b) isolating the effects among GVC firms, which Section 4.1 showed were the primary drivers of export growth.

The estimates are statistically indistinguishable from zero prior to the trade war across both panels. Following the increase in U.S. tariffs on Chinese goods, manufacturing output rises, with stronger effects among firms exporting to the United States, particularly after 2020.

In the full sample (Panel a of Figure 4a), the DD coefficient for U.S. exporters reaches 0.8 in 2021 and remains around 0.7 by 2023, with all post-shock coefficients precisely estimated. Among GVC firms (Panel b), the response is stronger. At the 2021 peak, the DD coefficients reach 0.7 for all exporters and 1.0 for U.S. exporters, corresponding to respective increases in manufacturing production of 13% and 19% at the 90–10 exposure difference.



(a) Full sample of Manufacturing Firms



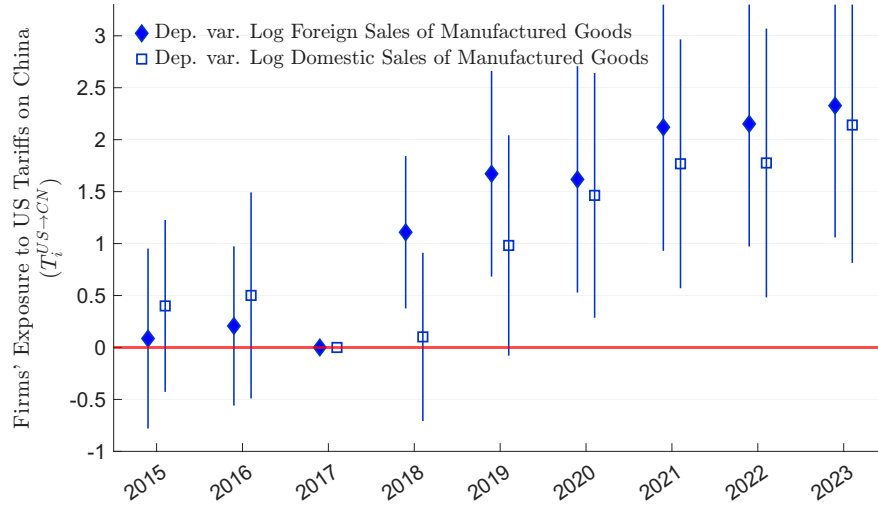
(b) GVC Firms (IMMEX)

Figure 4: Impact of US Tariffs on Chinese Goods on Firms' Production

Notes: The dependent variable is the natural logarithm of firms' value of production. Equation 3 is estimated using the treatment variable $T_i^{US \rightarrow CN}$ among 2017 exporters (Figure (a): $N = 51,838$ Figure (b): $N = 37,202$), and using the treatment variable $T_{i,US}^{US \rightarrow CN}$ among 2017 U.S. exporters (Figure (a): $N = 41,914$ Figure (b): $N = 33,418$). All regressions include firm fixed effects, three-digit industry-by-year fixed effects, and controls for concurrent trade policy changes. The y-axis displays yearly difference-in-differences coefficient estimates with 95% confidence intervals based on robust standard errors clustered at the firm level.



(a) Sales of Manufactured Goods at the Plant-level



(b) Foreign versus Domestic Sales of Manufactured Goods at the Plant-level

Figure 5: Impact of US Tariffs targeting China on Sales of Manufactured Goods

Notes: Estimation of Equation 4 using IMMEX plant-level data, N=19,410 (total sales), 15,967 (foreign), 17,804 (domestic). All regressions control for plant fixed effects, three-digit industry-by-year, municipality-by-year fixed effects and firm-level controls for concurrent trade policy changes. The y-axis displays the yearly DD coefficient estimates corresponding to $T_i^{US \rightarrow CN}$, along with 95% confidence intervals. Robust standard errors are clustered at the firm level.

4.2.1 Contract Manufacturing versus In-House Production

Establishments operating under IMMEX include both maquiladoras — export-processing plants that perform labor-intensive tasks on inputs supplied by U.S. clients and return the processed output under contract — and full-fledged manufacturing operations that produce goods within multinational firms’ own cross-border networks. The distinction matters for interpreting the implications for U.S. manufacturing. In contract manufacturing, the Mexican plant owns neither the inputs nor the output, performs limited transformation, and operates on a schedule dictated by U.S. clients’ manufacturing activity — expanding or contracting in lockstep with their production needs. This mode of production tends to serve smaller U.S. manufacturing firms outsourcing specific labor-intensive tasks (Antràs and Helpman 2004). In-house production, by contrast, involves the plant owning its inputs, undertaking more substantial transformation, and selling output within the firm’s network, with production decisions made independently of any single U.S. client’s schedule — and is more typically associated with larger multinationals that have internalized production across borders. As a result, contract manufacturing moves in tandem with U.S. manufacturing activity (Bergin et al. 2009), whereas in-house expansion can proceed independently and need not reflect changes in U.S. production.

Using the plant-level IMMEX data described in Section 2.2, I estimate Equation 4 with sales of own-manufactured goods—which exclude revenues from contract manufacturing—as the dependent variable to isolate in-house production. Figure 5 reports the results for total, foreign, and domestic sales separately. Figure 5a shows a strong positive response in total sales, with no evidence of differential pre-trends. By 2023, the DD coefficient reaches 1.92, corresponding to a 39% increase in own production at the 90–10 exposure difference. Figure 5b shows that the increase is driven primarily by foreign sales, for which the coefficient reaches 2.33 in 2023, corresponding to a 49% increase. Domestic sales also rise substantially, albeit with a one-year lag relative to foreign sales. This pattern is consistent with supply-chain deepening as production stages expand within Mexico.

To assess the role of contract manufacturing, I next examine total plant revenues, which include both sales of own production and revenues from contract work.¹⁵ Figure C.4 shows that while tariff exposure has a positive and statistically significant effect on total revenues, the response is substantially smaller than for own production. In 2023, the DD coefficient for foreign revenues is 0.87, compared with 2.33 for foreign sales of own production, and the coefficient for total revenues (0.67) is well below that for total own production (1.92).

Appendix Table C.2 reports estimates separately for own sales and contract manufacturing revenues, confirming a strong positive effect on the former and little to no effect on the latter.

Taken together, these results show that firms more exposed to U.S. tariffs on Chinese goods expand manufacturing output in Mexico, without a corresponding increase in export-processing (contract manufacturing) activity. Because contract manufacturing typically moves in tandem with U.S. production, this pattern suggests that the observed growth is not necessarily tied to U.S. manufacturing growth.

4.3 Within-Plant Adjustment: Material Inputs and Value Added

I next examine material inputs and value added to further characterize the nature of production expansion within plants.

Table C.3 shows that tariff exposure is associated with a large increase in imported (foreign) material inputs: the coefficient on log foreign materials is 1.48 (s.e. 0.39). Using the 17 percentage-point exposure gap between plants at the 90th and 10th percentiles, this implies roughly a 29% increase in imported material expenditures post-trade war. In contrast, domestic material inputs rise only modestly and imprecisely, while total materials increase by about 11% at the same exposure contrast. Plants scale export-oriented production primarily by expanding imported intermediate use, consistent with standard “importing to export” production in global value chains.¹⁶

¹⁵Since sales of own-manufactured goods exclude revenues from contract manufacturing services, plants that operate exclusively as contract manufacturers report zero own production and only *maquila* revenues. Accordingly, when total plant revenues are used as the outcome, the sample expands from 19,410 to 36,378 plants, reflecting the inclusion of establishments engaged solely in contract manufacturing.

¹⁶The modest increase in domestic material inputs is consistent with the one-year lag in domestic sales documented

At the same time, value added rises significantly (column 4). The estimated effect on total value added (0.42) implies a 7% increase at the 90–10 exposure contrast and the effect on domestic value added embodied in foreign sales (0.61) implies an 11% increase (column 5).¹⁷ Together, these results show that U.S. tariff exposure expands real production activity within Mexico, combining imported intermediates with domestic labor and other domestic factors to generate additional value embedded in export sales. However, the composition of this expansion indicates that it is strongly dependent on foreign inputs and firm organization, with domestic contributions arising primarily through labor rather than through upstream input production — the labor dimension is examined in detail next.

4.4 Effects of U.S. Tariffs Targeting China on Manufacturing Jobs in Mexico

This section examines how the shift in U.S. trade policy against China affected manufacturing employment in neighboring Mexico, a key production location within North American value chains. While the tariffs were promoted to ‘bring back jobs to America’, the analysis shows that instead of bringing jobs back home, they led to a significant increase in manufacturing employment in Mexico.

While higher output suggests increased labor demand, the link between output and employment is not mechanical, as firms may adjust through productivity improvements or the adoption of automation-intensive technologies (Diao et al. 2025; Firooz et al. 2025). To examine the employment response at the firm level, I estimate Equation 3 using the logarithm of the firm’s total employment across its manufacturing establishments.

Figure 6 presents results using the broad exposure measure and the U.S.-specific exposure measure, in Section 4.2.1 — both patterns suggest that the expansion within Mexico broadens over time but remains driven by foreign inputs and firm-internal networks rather than upstream domestic sourcing.

¹⁷Domestic value added in exports is constructed by netting foreign material and packaging expenses from foreign sales, plus foreign contract manufacturing revenues. Since this assumes all foreign inputs are used in production destined for the foreign markets, the measure understates domestic value added embodied in exports to the extent that some foreign inputs are used for domestic sales.

applied to 2017 exporters and 2017 U.S. exporters, respectively. The estimates show no evidence of differential pre-trends and a significant positive effect of U.S. tariff hikes on manufacturing employment under both exposure measures. Employment effects begin to emerge in 2019 and grow steadily through 2023, lagging output as firms expand production before adjusting their workforce.

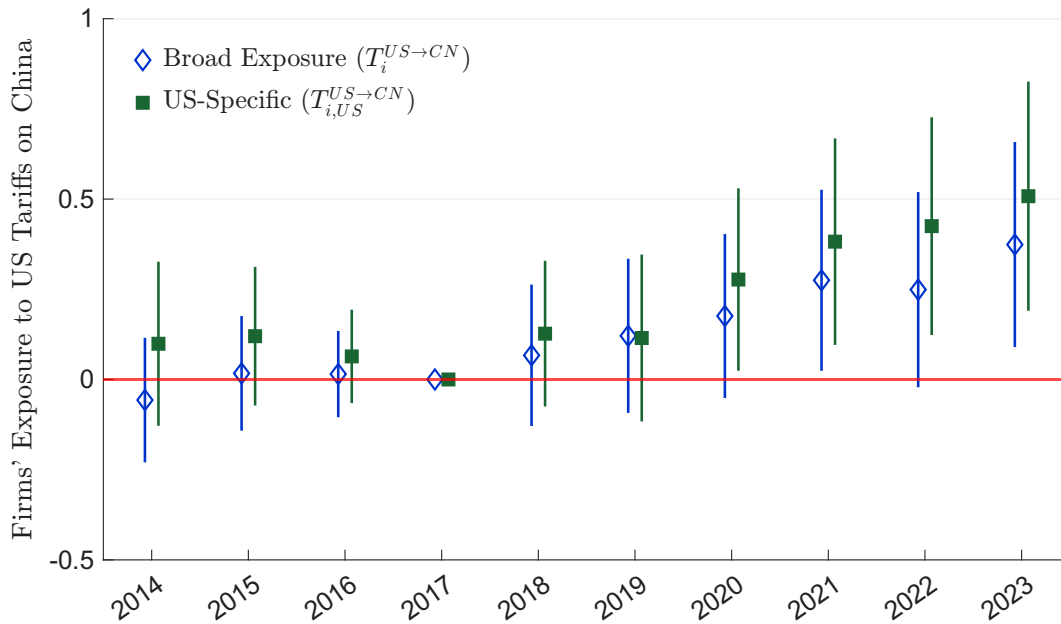


Figure 6: Impact of US Tariffs on Chinese Goods on Manufacturing Employment

Notes: Estimation of Equation 3. The dependent variable is the natural logarithm of firms' employment. 2017 Exporters, with $N=52,174$ and 2017 US Exporters, with $N=42,171$. All regressions control for firm fixed effects, three-digit industry-by-year fixed effects and concurrent trade policy changes. Robust standard errors are clustered at the firm level. The y-axis displays the yearly DD coefficient estimates corresponding to $T_i^{US \rightarrow CN}$ and $T_{i,US}^{US \rightarrow CN}$, along with 95% confidence intervals.

Table C.4 reports estimates separately for GVC and non-GVC firms and shows that employment gains are concentrated among GVC firms. By 2023, the DD coefficients for GVC firms reach 0.52 under the broad exposure measure and 0.57 under the U.S.-specific measure, both highly significant. The latter implies a 10% increase in manufacturing employment at the 90–10 exposure difference.

I next estimate the employment effect separately across two-digit industries. Figure 7 shows that employment gains are concentrated in technology-intensive industries, materials and chemicals (NAICS 32) and machinery and equipment manufacturing (NAICS 33)—with the statistically sig-

nificant effect in machinery and equipment emerging immediately in 2018. In contrast, employment effects in food and textile manufacturing (NAICS 31) remain near zero throughout the period. This pattern reflects the sectoral distribution of foreign multinationals integrated into North American production networks in Mexico, which are disproportionately concentrated in materials and machinery industries (Figure A.2).

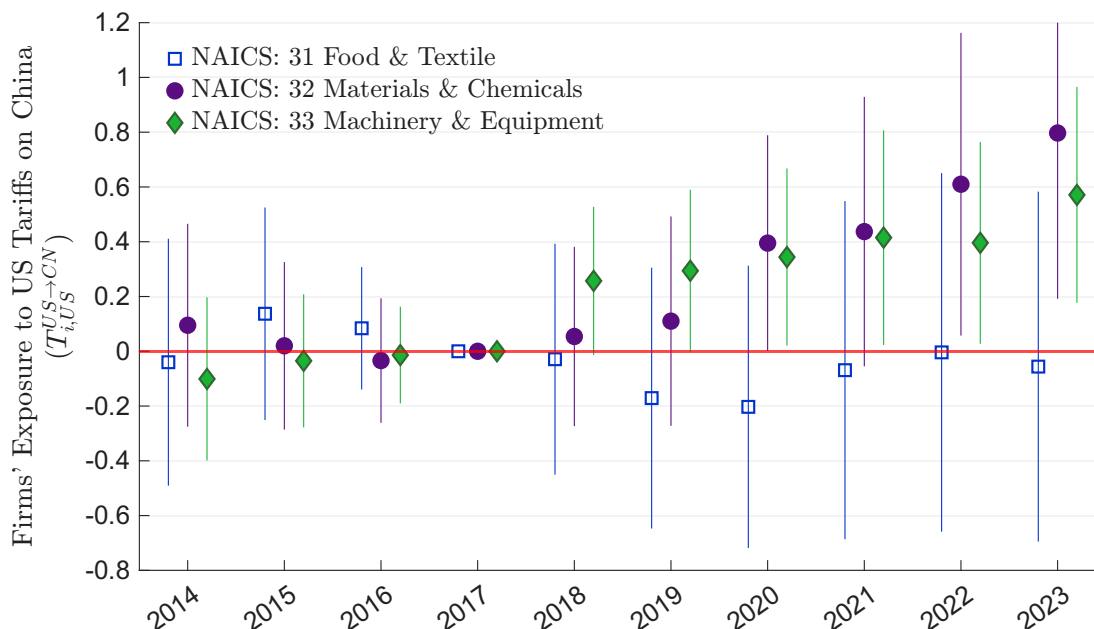


Figure 7: Employment gains are concentrated in tech-intensive industries

Notes: The sample consists of firms exporting to the United States as of 2017 in NAICS 31 (N = 8,197), NAICS 32 (N = 10,243), and NAICS 33 (N = 23,720). The dependent variable is the natural logarithm of firm employment. All regressions include for firm and year fixed effects and controls for concurrent trade policy changes. The y-axis displays the yearly DD coefficient estimates corresponding to $T_{i,US}^{US-CN}$, along with 95% confidence intervals based on robust standard errors clustered at the firm level.

Because firm-level measures aggregate activity across establishments, they may understate responses among multi-plant firms — through consolidation of activity toward tariff-exposed plants, reallocation across existing establishments, or expansion into new plants that attract resources from existing ones before appearing in the data. Appendix Figure C.5 addresses this by focusing on single-plant firms as of 2017, where none of these channels operate. The results show similar and stronger increases in both production and employment, suggesting the baseline estimates are attenuated among multi-plant firms.

I next examine how this employment growth is distributed across worker types within plants.

4.5 Within Plant Adjustment: Workforce Composition

To examine within-plant adjustment in greater depth, I turn to the IMMEX plant-level data. I use the occupation breakdown in the data to assess whether employment gains reflect expansion in core manufacturing activity.

Figure 8 presents estimates from Equation 4 for production workers and technicians and for non-production workers. The results show that exposure to U.S. tariff hikes leads to a significant and sustained increase in employment of production workers and technicians, beginning in 2018 and rising through 2023. In contrast, non-production employment exhibits much smaller and less precisely estimated responses, with coefficients close to zero and wider confidence intervals.

The increasing effects on production workers and technicians through 2023 point to a persistent expansion in production capacity rather than a transitory adjustment, underpinned by the value added increases documented in Section 4.3. Quantitatively, the 2023 coefficient of 0.7, combined with the 17 percentage-point difference in tariff exposure between the 90th and 10th percentiles, implies approximately a 13% increase in blue-collar employment at the plant level.

The employment measures in Figure 8 include both payroll and non-payroll workers. Because the plant-level data separately report payroll employment and workers supplied by third-parties, I test whether tariff-induced expansion initially relied on non-payroll arrangements, which were a low-cost and flexible hiring margin prior to the 2021 reform. Figure C.7 in the Appendix shows that in the early years of the trade war, tariff-induced employment gains were concentrated among non-payroll workers. This pattern is consistent with firms adjusting along more flexible and lower-cost labor margins in the presence of trade policy uncertainty (Handley and Limão, 2022).

From 2021 onward, the composition shifts markedly: payroll employment effects become large and statistically significant, while non-payroll employment effects diminish. Notably, non-payroll employment begins to decline even before the 2021 prohibition on the use of shell companies

for core business activities. This pattern suggests that, as tariffs became more clearly permanent and non-payroll arrangements less advantageous, firms transitioned from temporary to formal employment relationships, reflecting a shift from cautious, reversible expansion to more committed capacity growth.¹⁸

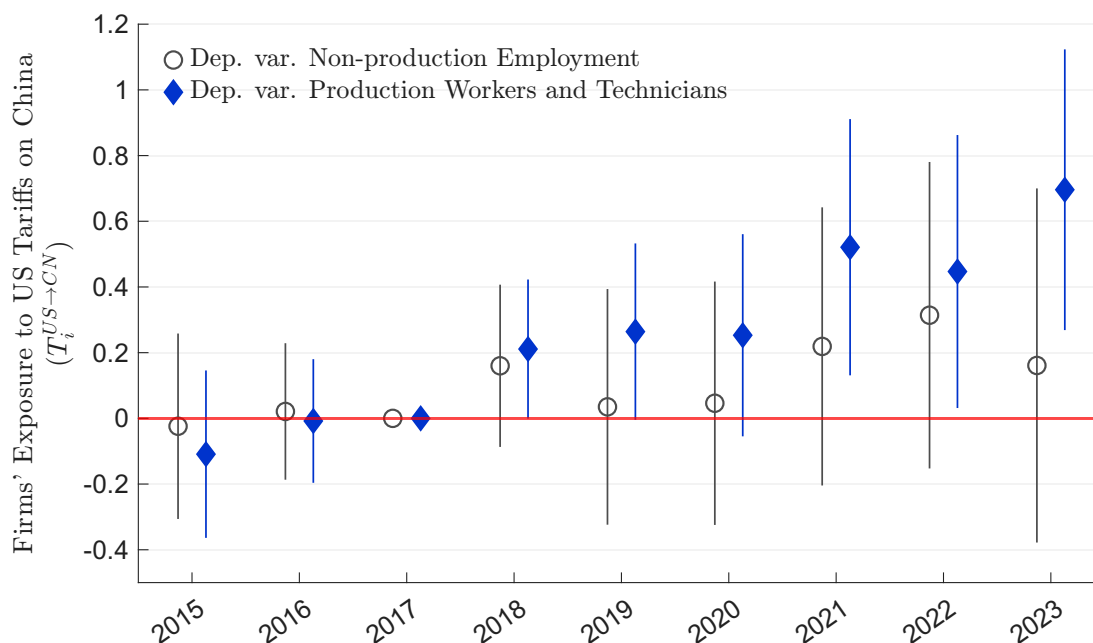


Figure 8: Tariff-Induced Employment Growth Concentrated in Production Occupations

Notes: Estimation of Equation 4 using IMMEX plant-level data, N=35,870 (non-production employment), N=36,336 (production workers and technicians). All regressions control for plant fixed effects, three-digit industry-by-year, municipality-by-year fixed effects and firm-level controls for concurrent trade policy changes. The y-axis displays the yearly DD coefficient estimates corresponding to $T_i^{US \rightarrow CN}$, along with 95% confidence intervals. Robust standard errors are clustered at the firm level.

4.6 Firm-Level Evidence on New Plant Openings

Beyond expansion at existing plants, a natural question is whether firms committed to longer-term capacity growth in Mexico by establishing new manufacturing facilities. While the data

¹⁸Estimated effects on wages are positive and statistically insignificant (Appendix Table C.5), suggesting that increased labor demand was absorbed through employment expansion without meaningful upward wage pressure relative to other plants in the same local labor market and year. This stands in contrast to Chen et al. (2025), who document declining average wages at more exposed firms using IMSS data through 2019. Since early expansion relied disproportionately on non-payroll labor — which falls outside IMSS payroll reporting — tariff-induced labor demand was not fully captured in IMSS records, helping reconcile the two findings.

do not directly measure capital investment, new plant openings serve as a meaningful indicator of long-term capacity expansion, reflecting decisions to add production lines and infrastructure. Indeed, since 2018, major multinationals—including Bosch, LEGO, Continental, Volkswagen, and Toyota, all operating under Mexico’s IMMEX program—have announced plans to expand their Mexican manufacturing footprint. A large share of these projects involves the construction of new manufacturing plants (News Desk 2018; Webwire 2022; Reuters 2022a; Cohen et al. 2025).

Figure C.8 in the Appendix reports the estimated effect of the U.S. trade policy shift on the number of manufacturing plants operated by firms. The results show a clear shift after 2018, consistent with a reallocation of production capacity toward Mexico in response to the U.S.–China trade conflict. Because the outcome in Figure C.8 is the log number of plants, the effect captures the net entry margin, plant openings minus closures, rather than openings alone.

To isolate the effect on plant entry, I identify newly established plants using the IMMEX registry over the period 2016–2024. A plant is classified as new if it appears for the first time in the plant registry under firm i at an address not previously associated with that firm. For each firm i and year t , I construct the total number of newly opened plants, which serves as the outcome variable in the entry regressions.

To maintain comparability with the main analysis, I restrict the IMMEX registry sample to manufacturing firms active as of 2017 and estimate a firm-level specification in which tariff exposure is defined at the three-digit industry level, constructed from the firm-level trade data analogously to the firm-level measures.¹⁹ The specification includes firm and border-by-year fixed effects and controls for concurrent trade policy exposures.

The results presented in Table 2 show that U.S. tariffs on China spurred the establishment of new manufacturing plants in Mexico. The dependent variable is the inverse hyperbolic sine transformation of the number of new plants per firm per year. Column (1) shows that across all firms, greater tariff exposure is associated with significantly more plant openings, with a coefficient of

¹⁹The IMMEX registry is maintained by the Ministry of Economy and falls outside INEGI’s confidential microdata environment; this analysis is therefore conducted outside INEGI’s secure servers, which necessitates industry-level rather than firm-level exposure measures.

0.59. Columns (2) and (3) split the sample by firm nationality and reveal that the effect is 2.5 times larger among foreign-owned firms (0.791) than for non-foreign firms (0.317) and both statistically significant. Moving from the 10th to the 90th percentile of tariff exposure (0.10–0.23) is associated with 0.106 additional new plants per firm per year among foreign firms — a 64% increase relative to their mean of 0.166 — and 0.041 additional new plants per firm per year among non-foreign firms, a 53% increase relative to their mean of 0.077.²⁰

Columns (4)–(6) augment the specification with an interaction between tariff exposure and Chinese input dependence. Chinese input dependence is measured by $IMCN_j^{US \rightarrow CN}$, which captures the share of industry j 's imports in 2017 that originated from Chinese products subject to US tariffs on China, weighted by the magnitude of those tariff increases (Equation B.4). I construct this measure at the three-digit industry level by aggregating firm-level import data, and define the indicator $\mathbf{1} \left[IMCN_j^{US \rightarrow CN} \geq \text{Median} \right]$ accordingly.

The interaction term is large and significant for foreign firms (2.8) but small and insignificant for non-foreign firms (-0.2), suggesting that the plant-opening response among foreign firms is concentrated in those that rely more heavily on Chinese inputs — consistent with a supply chain reorganization motive. The baseline tariff exposure, $T_j^{US \rightarrow CN}$ already captures the degree to which a Mexican firm's substitute for Chinese goods in the US market, so trade diversion alone should show up in the baseline coefficient. Columns (7) further distinguishes by US-owned from other foreign-owned firms. The interaction with Chinese input dependence is even larger at 4.9, nearly double the pooled foreign estimate, suggesting that the non-linear effect is driven by the US-owned firms. For US-owned firms above the median of Chinese input dependence, moving from the 10th to the 90th percentile of tariff exposure is associated with 0.676 additional new plants per firm per year, nearly four times their mean, implying 338 additional new plants per 1,000 US-owned firms per year. The additional role of the interaction term points to a more specific mechanism: Mexican

²⁰These figures are obtained by inverting the inverse hyperbolic sine transformation. Specifically, for a coefficient $\hat{\beta}$ and a change in tariff exposure ΔT (the difference between the 90th and 10th percentiles), the predicted change in the outcome in levels is $\sinh(\text{IHS}(\bar{y}) + \hat{\beta} \times \Delta T) - \bar{y}$, where $\bar{y}$ is the sample mean of new plants per firm per year and $\sinh(x) = (e^x - e^{-x})/2$ is the inverse of the IHS transformation.

affiliates of US firms that operate in industries which both compete with Chinese goods in the US market and depend more on Chinese inputs face the strongest incentive to expand and respond by opening new plants.

Column (8) explores heterogeneity by Chinese ownership, combining firms headquartered in China, Hong Kong, and Singapore due to their small individual counts. Despite their limited aggregate presence, the significant interaction (1.963) together with the positive and significant baseline effect suggests that Chinese-owned affiliates respond disproportionately to tariff exposure, providing empirical evidence on the behavior of Chinese-owned firms that has featured prominently in public debate.²¹

4.7 The Role of U.S.-headquartered Firms in Creating Jobs in Mexico

U.S.-headquartered manufacturing firms in IMMEX accounted for 34% of Mexico’s manufacturing exports to the United States and nearly one million manufacturing jobs in 2017.²² Given the central role of U.S. multinationals in North American production networks (Flaaen et al. 2025), and evidence that Mexico’s export surge between 2018 and 2021 was driven by foreign firms embedded in global value chains (Utar et al. 2025), U.S.-headquartered firms provide a useful setting to examine how U.S. trade policy reshaped regional production. This section examines how these firms adjusted their exports, sourcing, production and employment decisions.

The rise in new manufacturing plants by U.S.-headquartered firms in Mexico—documented in Section 4.6—is particularly notable given the absence of corresponding gains in U.S. manufacturing output or employment (Flaaen and Pierce, 2024; Autor et al., 2024). The new plant entry by U.S. firms occurs in industries that both rely on Chinese inputs and compete with Chinese suppliers. This raises the question of whether higher input costs from U.S. tariffs discouraged domestic assembly and redirected production toward Mexico.

²¹Appendix Table D.1 replicates the analysis using the continuous $IMCN_j^{US \rightarrow CN}$ measure instead of the binary indicator and reports wild cluster bootstrap p-values to address possible concerns about standard error reliability with a small number of industry-level clusters. The results are qualitatively unchanged. Appendix Table D.2 presents PPML estimates using the count of new plants directly as the outcome, with broadly consistent results.

²²Author’s calculation based on manufacturing firms in PEME linked to the IMMEX registry.

Table 2: Rising US import tariffs on China spur plant openings in Mexico

	Linear			Linear $\times$ CN Input Dep.			Parent (8) Foreign	
	(1) All	(2) Foreign	(3) Not Foreign	(4) All	(5) Foreign	(6) Not Foreign		(7) US
$\mathbf{T}_j^{\text{US} \rightarrow \text{CN}} \times \text{PostTW}_t$	0.590*** (0.194)	0.791** (0.368)	0.317*** (0.092)	0.194 (0.239)	0.076 (0.310)	0.293 (0.176)	-0.296 (0.339)	0.723* (0.375)
$\mathbf{T}_j^{\text{US} \rightarrow \text{CN}} \times \text{PostTW}_t \times \text{CN Input Dep.}$				1.371* (0.786)	2.822*** (0.714)	-0.226 (0.526)	4.881*** (0.974)	
$\mathbf{T}_j^{\text{US} \rightarrow \text{CN}} \times \text{PostTW}_t \times \text{CN Affiliate}$								1.963*** (0.628)
Observations	33,627	17,166	16,461	33,619	17,166	16,453	10,739	17,193
R-squared	0.288	0.284	0.283	0.289	0.285	0.283	0.299	0.285
Concurrent Trade Policy Controls	✓	✓	✓	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓	✓	✓	✓
Border $\times$ Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Chinese Input Dependence $\times$ PostTW _{<i>t</i>}	–	–	–	✓	✓	✓	✓	–
CN Affiliate $\times$ PostTW _{<i>t</i>}	–	–	–	–	–	–	–	✓
Avg. Number of New Plants	0.122	0.166	0.077	0.122	0.166	0.077	0.178	0.166

Notes: Manufacturing IMMEX firms operating as of 2017, over 2017–2024. The dependent variable is the inverse hyperbolic sine transformation of the number of new plants. Concurrent Trade Policy Controls include exposures to the U.S.–Mexico tariff escalation, China’s retaliatory tariffs, and the input channel measures $IMU S_j^{\text{US} \rightarrow \text{CN}}$ and $IMU S_j^{\text{US} \rightarrow \text{CN}}$, all constructed using firm-level data aggregated to the industry level and interacted with the post–trade war indicator. Columns (4)–(7) further control for Chinese Input Dependence as defined by $\mathbf{1} [\text{IMCN}_j^{\text{US} \rightarrow \text{CN}} \geq \text{Median}]$. CN Affiliate is a dummy variable that takes the value of one if the parent company headquarters are located in China, Hong Kong, or Singapore. All specifications include firm and border $\times$ year fixed effects. Robust standard errors are clustered at the three-digit industry level. ***, **, *, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Figure 9a plots the export response of U.S.-headquartered manufacturing firms in Mexico to rising U.S. tariffs on China. The estimates show no evidence of pre-trends, with coefficients close to zero through 2017. Following the tariff increases, exports rise gradually through 2019, dip in 2020 during the COVID-19 pandemic, and strengthen markedly from 2021 onward. The estimate reaches 1.84 by 2023, similar in magnitude to the full-sample estimate, implying a 37% increase in exports between firms at the 90th and 10th percentiles of tariff exposure.

Figure 9b traces the year-by-year response of U.S.-headquartered firms' imports from the United States and China. Imports from the United States rise alongside exports, reflecting the structure of multinational production networks in which affiliates source inputs from their headquarters country. Imports from China also rise, but with a more delayed response, becoming more pronounced from around 2020 onward. This sequencing suggests that the tariff-induced expansion of U.S. firms in Mexico initially relied more heavily on existing North American supply chains, with Chinese sourcing increasing as Mexican operations scaled up.²³

A possible mechanism is that higher tariffs on Chinese inputs increased the cost of final assembly in the United States, leading U.S. firms to shift downstream production stages to Mexican affiliates. These plants increasingly combined inputs from both the United States and China while exporting final goods to the U.S. market. Consistent with this interpretation, I find that U.S. tariff exposure significantly increases the share of consumption goods in U.S. firms' exports, with the effect emerging from 2020 onward (Figure C.9). This timing aligns with the increase in Chinese input sourcing, suggesting that as Mexican operations scaled up, they increasingly assembled final goods combining inputs from both the United States and China.

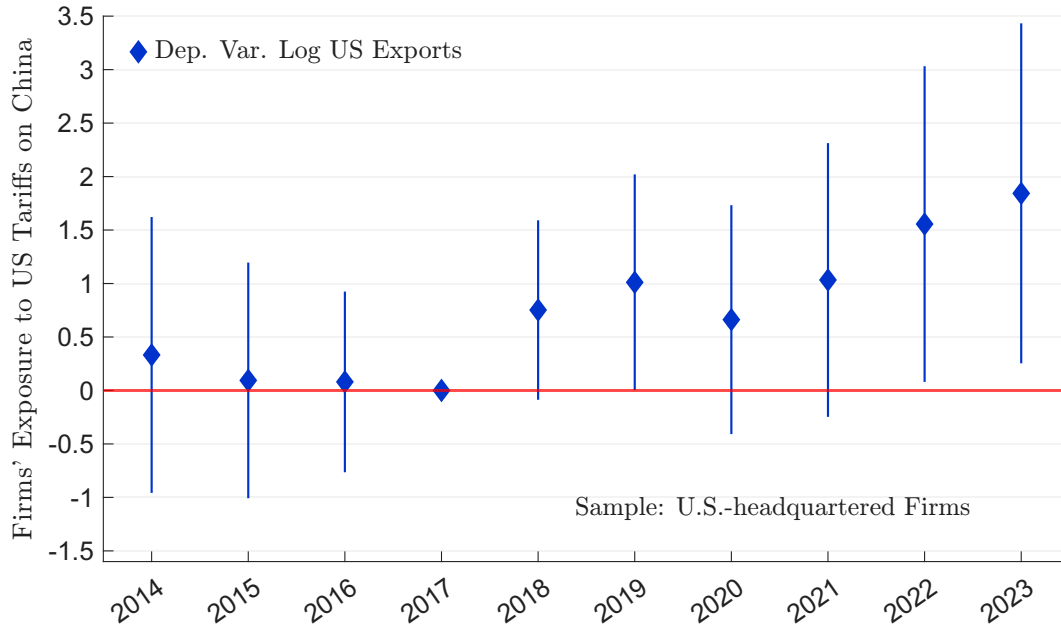
Notably, this pattern of deepening sourcing from both the United States and China appears specific to U.S.-headquartered firms. Table C.6 in the Appendix shows that while both U.S.- and non-U.S. foreign firms expand exports to the United States, only U.S.-headquartered firms significantly increase imports from both the United States and China. If anything, the share of imports from

²³The initial reliance on existing North American supply chains is consistent with high switching costs in buyer-supplier relationships (Monarch 2022), which discourage firms from replacing established partners even when tariffs alter relative costs.

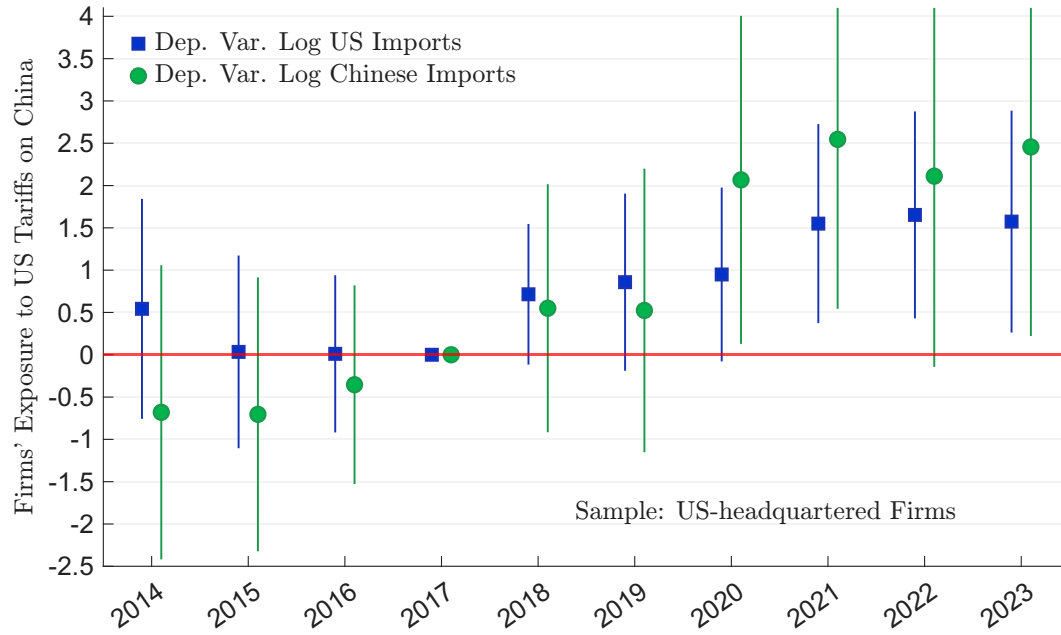
China declines among other foreign firms as tariff exposure rises (Figure C.10), suggesting that their Mexican operations substituted more directly for Chinese production.

Table 3 reports estimates of Equation 3 for U.S.-headquartered manufacturing firms. Higher U.S. tariffs on China increase production among U.S. firms in Mexico and are followed by later gains in employment, alongside positive effects on labor productivity in several post-2018 years. The joint rise in output, employment, productivity, and plant entry indicates that U.S.-headquartered firms expanded longer-term manufacturing capacity in Mexico, rather than diverting trade through minimally transformed goods, and played a significant role in the U.S. trade policy-induced manufacturing expansion in Mexico.

Taken together, these findings reveal a striking pattern: the very firms that were incentivized by the U.S. trade policy to expand domestically responded instead by deepening their manufacturing operations in Mexico — increasing exports to the United States, scaling up sourcing from both the United States and China, raising output and employment, and establishing new plants. This provides new firm-level evidence on how U.S. multinationals reallocated production within their own cross-border networks in response to trade policy.



(a) Export Responses of U.S.-Headquartered Firms



(b) Import Responses of U.S.-Headquartered Firms

Figure 9: Trade Responses of U.S.-Headquartered Firms to U.S. Tariffs on China

Notes: Top panel shows the response of U.S. exports by U.S.-headquartered firms in Mexico (N=10,940). Bottom panel shows import responses from the United States (N=10,955) and China (N=10,459). Estimation of Equation 3. All regressions include firm fixed effects, three-digit-industry-by-year fixed effects, and controls for concurrent trade policy changes. The y-axis reports yearly DD coefficient estimates corresponding to $T_i^{US \rightarrow CN}$, with 95% confidence intervals.

Table 3: U.S. Tariffs on China and Expansion of U.S.-Headquartered Firms in Mexico

	(1) Log Production	(2) Log Employment	(3) Log Labor Productivity
$T_{i,US}^{US \rightarrow CN} \times 2014$	0.049 (0.332)	0.348 (0.310)	0.001 (0.245)
$T_{i,US}^{US \rightarrow CN} \times 2015$	-0.156 (0.278)	0.194 (0.257)	-0.103 (0.202)
$T_{i,US}^{US \rightarrow CN} \times 2016$	-0.078 (0.182)	0.084 (0.134)	-0.080 (0.151)
$T_{i,US}^{US \rightarrow CN} \times 2018$	0.714** (0.338)	0.296 (0.237)	0.466* (0.269)
$T_{i,US}^{US \rightarrow CN} \times 2019$	0.419 (0.364)	0.127 (0.249)	0.370 (0.291)
$T_{i,US}^{US \rightarrow CN} \times 2020$	0.517 (0.382)	0.267 (0.264)	0.323 (0.302)
$T_{i,US}^{US \rightarrow CN} \times 2021$	1.302*** (0.449)	0.551* (0.293)	0.945** (0.395)
$T_{i,US}^{US \rightarrow CN} \times 2022$	1.134*** (0.411)	0.510* (0.306)	0.725** (0.307)
$T_{i,US}^{US \rightarrow CN} \times 2023$	0.772 (0.473)	0.602* (0.349)	0.293 (0.414)
N	10,522	10,553	10,521
Firm FE	✓	✓	✓
3-digit Industry $\times$ Year FE	✓	✓	✓
Concurrent Trade Policy Controls	✓	✓	✓

Notes: Sample: U.S.-headquartered firms (with global ultimate headquarters in the United States) that exported to the United States as of 2017. Concurrent Trade Policy Controls include firms' exposures to the US-Mexico tariff escalation, retaliatory tariffs of China and the input channels as described in equations B.1-B.6. Robust standard errors are clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

5 Concluding Remarks

The 2018–2019 U.S.–China trade war marked one of the most dramatic shifts in U.S. trade policy in decades. While the tariffs, estimated to impose large costs on U.S. consumers and producers, were promoted as a means of revitalizing domestic manufacturing, this paper shows that they led to a significant manufacturing expansion in Mexico—the United States’ key production partner within North American supply chains.

The findings show that the trade war significantly increased manufacturing output, value added, employment, and the number of plants in Mexico. These gains are driven by foreign multinationals and their domestic affiliates operating under Mexico’s export platform, IMMEX. The tariffs induce growth in in-house production rather than contract manufacturing (*maquila*), with value added and employment rising in technology-intensive industries—concentrated among production workers and technicians, the very segment of the workforce that trade protectionism aimed to restore domestically. The expansion ran through pre-existing North American production networks, highlighting the important role of existing MNE networks in shaping the transmission of trade policy shocks.

U.S.-headquartered firms were at the center of this expansion, with their Mexican subsidiaries significantly increasing production, employment, and establishing new plants in response to the tariffs. This pattern suggests that the adjustment to U.S. protectionism occurred through the reallocation of activity within multinational networks. The evidence documented in this paper also raises the possibility that expansion in Mexico did not come only in expense of Chinese operations but also through the relocation of downstream production stages away from the United States. Combined with the absence of any measurable expansion in U.S. manufacturing output or employment (Flaen and Pierce 2024; Autor et al. 2024), this points to a striking disconnect between the intended and realized effects of trade protection: the very firms the policy sought to incentivize to expand domestically responded instead by deepening their manufacturing operations abroad.

What does this mean for Mexico? While U.S. tariffs delivered a significant boost to manufac-

turing activity, its character warrants attention. Production gains are concentrated among foreign multinationals responding to a global trade policy shock originating outside Mexico, with limited effects on domestic firms outside the IMMEX program. The expansion relies on imported intermediates with domestic labor rather than reflecting the emergence of a broader domestic industrial base, suggesting that the benefits of deep integration into North American supply chains, while real, remain contingent on external policy decisions and unevenly distributed across firms.

Overall, the findings suggest that tariff policy may be ill-suited to achieving domestic manufacturing objectives in a deeply integrated regional economy, where production is organized through global value chains. Rather than helping manufacturing at home, protectionist trade policy measures may reinforce cross-border production linkages, shifting manufacturing activity to proximate economies while leaving domestic manufacturing outcomes largely unchanged.

References

- Abadie, A., S. Athey, G. W. Imbens, and J. Wooldridge** 2023. “When should you adjust standard errors for clustering?”, *Quarterly Journal of Economics*, Vol 138(1):1-35.
- Alfaro, L., and D. Chor** 2023. “Global Supply Chains: The Looming of Great Reallocation”, NBER Working Paper No. 31661.
- Amiti, M., S. J. Redding, and D. E. Weinstein** 2019. “The Impact of the 2018 Tariffs on Prices and Welfare.” *The Journal of Economic Perspectives*, 33(4), 187–210.
- Amiti, M., S. J. Redding, and D. E. Weinstein** 2020. “Who’s Paying for the US Tariffs? A Longer-Term Perspective.” *AEA Papers and Proceedings*, 110: 541-46.
- Antràs, P. and D. Chor** 2022. “Global Value Chains”, *Handbook of International Economics* Vol. 5. Elsevier.
- Antràs, P., E. Fadeev, T. Fort, and F. Tintelnot** 2022. “Global Sourcing and Multinational Activity: A Unified Approach”, NBER WP 30450.
- Antràs, P. and E. Helpman** 2004. “Global Sourcing.” *Journal of Political Economy*, 112(3): 552–580.
- Autor, D., A. Beck, D. Dorn, and G. Hanson** 2024. “Help for the Heartland? The Employment and Electoral Effects of the Trump Tariffs in the United States”, NBER WP 32082.
- Beason, R., R. Liu, P. T. Nguyen, and L. Wang** 2025. “The Third-Country Impact of the US-China Trade War on Vietnamese Firms: A Global Value Chains Perspective,” Working Paper, SSRN 5113547.
- Bergin, P. R., R. C. Feenstra, and G. H. Hanson** 2009. “Offshoring and Volatility: Evidence from Mexico’s Maquiladora Industry”, *American Economic Review*, Vol. 99, No. 4, pp. 1664–1671.
- Blanchard, E. J., C. P. Bown, and R. C. Johnson** 2026. “Global Value Chains and Trade Policy”, *Review of Economic Studies*, Vol. 93, Issue 1, pp. 181–214.
- Bown, C.** 2021. “The US-China Trade War and Phase One Agreement”, *Journal of Policy Modeling*, 43:805–843.
- Cavallo, A., G. Gopinath, B. Neiman, and J. Tang.** 2021. “Tariff Pass-Through at the Border and at the Store: Evidence from US Trade Policy.”, *American Economic Review: Insights*, 3 (1): 19-34.
- Chen, N., D. Novy, and D. Solarzano.** 2025. “Trade Diversion and Labor Market Outcomes”, CEPR Discussion Paper No. 20296.
- Cohen, P., R. Akira, and E. Nelson.** 2025. “High Stakes for Global Companies in Trump’s Latest Tariffs.” *New York Times*, January 31 2025.
- Conconi, P., F. Leone, G. Magerman, and C. Thomas** (2025). “Multinational Networks and Trade Participation,” *Discussion Paper Series, Department of Economics, University of Oxford*, DP No. 1071, March 2025.
- Diao, X., M. Ellis, M. McMillan, and D. Rodrik.** 2025. “Africa’s Manufacturing Puzzle: Evidence from Tanzanian and Ethiopian Firms”, *The World Bank Economic Review*, 39(2): 308-340.

- Estefan, A., R. Gerhard, J. P. Kaboski, I. O. Kondo, and W. Qian** 2024. “Outsourcing Policy and Worker Outcomes: Causal Evidence from a Mexican Ban,” NBER Working Paper 32024.
- Fajgelbaum, P. D., P. K. Goldberg, P. J. Kennedy, and A. K. Khandelwal** 2020. “The Return to Protectionism”, *The Quarterly Journal of Economics*, 135, 1–55.
- Fajgelbaum, P., and Khandelwal, A.** 2022. “The Economic Impacts of the US-China Trade War”, *Annu. Rev. Econ.* vol.14 DOI: 10.1146/annurev-economics-051420-110410.
- Fajgelbaum, P. D., P. K. Goldberg, P. J. Kennedy, A. K. Khandelwal, and D. Taglioni** 2024. “Trade War and Global Reallocations”, *American Economic Review: Insights*, 6(2):295–312.
- Flaen, A. and J. R. Pierce** 2024. “Disentangling the Effects of the 2018-2019 Tariffs on a Globally Connected U.S. Manufacturing Sector”, *The Review of Economics and Statistics*.
- Flaen, A., A. Hortaçsu, and F. Tintelnot** 2020. “The production relocation and price effects of US trade policy: the case of washing machines”, *American Economic Review*, 110:2103–27.
- Flaen, A. B., F. Kamal, E. Lee, and K-M. Yi** 2025. “An Anatomy of U.S. Establishments’ Trade Linkages in Global Value Chains,” NBER Working Paper 33724.
- Firooz, H., S. Leduc, and Z. Liu** 2025. “Reshoring, automation, and labor markets under trade uncertainty”, *Journal of International Economics*, 104, Article 104091.
- Freund, C., A. Mattoo, A. Mulabdic, and M. Ruta** 2024. “Is US trade policy reshaping global supply chains?”, *Journal of International Economics*, 152, Article 104011.
- Garred, J. and S. Yuan** 2025. “Relocation from China (with Chinese Characteristics)”, *Journal of Development Economics*, 176.
- Glickman, Ben** 2024. “Stellantis to Add Mexican Output — Move to build more Rams in country comes amid union criticism of job cuts”, Wall Street Journal, Eastern edition; New York, N.Y., Oct 16, 2024
- Gopinath, G., P. Gourinchas, A. Presbitero, and P. Topalova.** 2025. “Changing global linkages: A new Cold War?”, *Journal of International Economics*, 2025-01, Vol.153, Article 104042.
- Graziano, A. G., M. Sztajerowska, and C. Volpe Martincus** 2024. “Trading Places: How Trade Policy Is Reshaping Multinational Firms’ Location”, *CESifo Working Paper* No. 11514.
- Grossman, G., E. Helpman, and S. Redding.** 2024. “When Tariffs Disrupt Global Supply Chains.”, *American Economic Review*, 114(4): 988–1029.
- Gutiérrez, A., S. Heise, N. Rizzotti, and F. Tintelnot** 2025. “Multinational Production and Trade Policy”, Working Paper.
- Handley, K., F. Kamal, and R. Monarch** 2025. “Rising Import Tariffs, Falling Export Growth: When Modern Supply Chains Meet Old-Style Protectionism”, *American Economic Journal: Applied Economics*, vol. 17, no. 1, pp.208–38.
- Handley, K., N. Limão** 2022. “Trade Policy Uncertainty”, *Annual Review Economics*, 14:363–395.
- McCaig, B., N. Pavcnik, and W. F. Wong** 2025. “Foreign and Domestic Firms: Long Run Employment Effects of Export Opportunities,” *The Review of Economics and Statistics*, forthcoming.

Monarch, R. 2022. “‘It’s Not You, It’s Me’: Prices, Quality, and Switching in U.S.-China Trade Relationships,” *The Review of Economics and Statistics*, MIT Press, vol. 104(5), pages 909-928, December.

News Desk 2018. “Bosch to Invest \$120 M for New Plant in Mexico.”, EE Times, May 1.

Reuters 2020. “Foxconn, Other Asian Firms Consider Mexico Factories as China Risks Grow.” August 24.

Reuters 2022a. “Volkswagen, Continental Announce Nearly \$1 bln in Mexico Investments.” October 27.

Reuters 2022b. “Deere to Move Cab Production to Mexico amid Tight U.S. Labor Market.” June 3.

Rotunno, L., S. Roy, A. Sakakibara, and P. Vézina. 2023. “Trade Policy and Jobs in Vietnam: The Unintended Consequences of Trump’s Trade War”, QPE Working Paper 2023-56.

The White House. 2018. “Made in America: Businesses and Workers Are Thriving Under President Donald J. Trump.” July 23. <https://trumpwhitehouse.archives.gov/briefings-statements/made-america-businesses-workers-thriving-president-donald-j-trump/>.

The White House. 2019. “The Historic Results of President Donald J. Trump’s First Two Years in Office.” January 20. <https://trumpwhitehouse.archives.gov/briefings-statements/the-historic-results-of-president-donald-j-trumps-first-two-years-in-office/>

Utar, H. 2025. “Firms and Labor in Times of Violence: Evidence from the Mexican Drug War”, *The World Bank Economic Review*, Volume 39, Issue 3, pp.632–662, <https://doi.org/10.1093/wber/lhae037>

Utar, H., and L. B. T. Ruiz 2013. “International Competition and Industrial Evolution: Evidence from the Impact of Chinese Competition on Mexican Maquiladoras”, *Journal of Development Economics*, Vol. 105, November 2013, 267–287.

Utar, H., A. Cebreros Zurita and L. Torres 2025. “The US-China Trade War and the Relocation of Global Value Chains to Mexico”, *Review of Economics and Statistics*, <https://doi.org/10.1162/REST.a.1682>.

Webwire. 2022. “LEGO Group Invests in Expansion of Factory in Monterrey.” August 5.

Appendix to:

“Elsewhere in North America”

Hâle Utar

Contents (Appendix)

A	Descriptive Statistics	1
B	Data and Variable Construction	4
B.1	Overview of Categories, Variables, and Classifications	5
B.2	Firm-Level Controls for Concurrent Trade Policy Changes	6
B.3	Sample Restriction for the U.S.-Specific Exposure Measure	8
C	Additional Analysis	10
C.1	Firms' Trade Outcomes	10
C.2	Supplemental Analysis—Manufacturing Output	14
C.3	Supplemental Analysis—Manufacturing Employment	17
C.4	Aggregation Across Establishments and Single-Plant Firms	18
C.5	Within-Plant Adjustment: Additional Results	21
C.6	Adjustment of U.S.-headquartered firms	24
D	Robustness Analysis	26
D.1	Sequential Inclusion of Controls	26
D.2	Robustness—Plant Openings in Mexico	30
D.3	Robustness to USMCA-Affected Industries	33

Appendix Figures

Figure A.1: Distribution of firms across industries

Figure A.2: Distribution of GVC firms across industries by ownership type

Figure C.1: Impact of US Tariffs Targeting China on Manufacturing Firms' Total Exports and Imports

Figure C.2: Impact of US Tariffs Targeting China on US Exports Net of Chinese Imports

Figure C.3: How US Exports of Foreign GVC Firms Respond to the US Tariffs

Figure C.4: Impact of the U.S. Tariff Hikes Targeting China on Plants' Revenue

Figure C.5: Employment and Production Effect among Single Plant Firms

Figure C.6: The Effect of Rising U.S. Import Tariffs on China on Labor Productivity

Figure C.7: Rising U.S. Tariffs on China and Labor Adjustment Margins

Figure C.8: Rising US import tariffs on China spur plant openings in Mexico

Figure C.9: US MNE Export Growth Tilts Toward Final Goods

Figure C.10: Import Share from China in Response to Rising US Tariffs: U.S.- vs. Other Foreign-Headquartered Firms

Figure D.1: U.S. tariffs on China and firms' exports to the United States, with sequentially added controls

Figure D.2: U.S. tariffs on China and firms' production, with sequentially added controls

Figure D.3: U.S. tariffs on China and firms' employment, with sequentially added controls

Figure D.4: Robustness to Excluding Firms Affected by USMCA Provisions

Appendix Tables

[Table A.1: Firm-Level Measures of Exposures to the Trade War](#)

[Table A.2: Summary Statistics—GVC Firms by Country of Ownership](#)

[Table B.1: U.S. Export Share Gains by Baseline U.S. Export Share](#)

[Table C.1: The US-China Trade War and Manufacturing Firms' Trade Outcomes](#)

[Table C.2: Impact of U.S. Tariffs Targeting China on Plant-Level Revenue and Sales](#)

[Table C.3: Impact of U.S. Tariffs Targeting China on Material Inputs and Value Added](#)

[Table C.4: The US-China Trade War and Manufacturing Employment](#)

[Table C.5: Impact of U.S. Tariffs Targeting China on Wages](#)

[Table C.6: Rising Chinese Imports in US MNEs in Mexico](#)

[Table D.1: Plant Openings in Mexico: Continuous Chinese Input Dependence and Wild Cluster](#)

[Bootstrap Inference](#)

[Table D.2: Plant openings in Mexico: PPML Estimates](#)

A Descriptive Statistics

Table A.1: Firm-Level Measures of Exposures to the Trade War

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Mean	SD	P10	P25	P75	P90	N
All Exporters							
$T_i^{\text{US} \rightarrow \text{CN}}$	0.201	0.074	0.077	0.152	0.250	0.250	52,260
$T_{i,\text{US}}^{\text{US} \rightarrow \text{CN}}$	0.150	0.095	0.000	0.056	0.242	0.250	52,260
$T_i^{\text{US} \rightarrow \text{MEX}}$	0.003	0.022	0.000	0.000	0.000	0.000	52,260
$T_i^{\text{MEX} \rightarrow \text{US}}$	0.004	0.019	0.000	0.000	0.000	0.004	52,260
$T_i^{\text{CN} \rightarrow \text{US}}$	0.167	0.072	0.058	0.110	0.225	0.250	52,260
$IMUS_i^{\text{US} \rightarrow \text{CN}}$	0.103	0.084	0.000	0.019	0.175	0.224	52,260
$IMCN_i^{\text{US} \rightarrow \text{CN}}$	0.026	0.046	0.000	0.000	0.030	0.081	52,260
$IMUS_i^{\text{CN} \rightarrow \text{US}}$	0.081	0.071	0.000	0.015	0.132	0.183	52,260
GVC Firms							
$T_i^{\text{US} \rightarrow \text{CN}}$	0.203	0.072	0.087	0.157	0.250	0.250	37,440
$T_{i,\text{US}}^{\text{US} \rightarrow \text{CN}}$	0.169	0.086	0.014	0.117	0.245	0.250	37,440
$T_i^{\text{US} \rightarrow \text{MEX}}$	0.003	0.025	0.000	0.000	0.000	0.000	37,440
$T_i^{\text{MEX} \rightarrow \text{US}}$	0.005	0.021	0.000	0.000	0.000	0.006	37,440
$T_i^{\text{CN} \rightarrow \text{US}}$	0.166	0.072	0.059	0.108	0.225	0.249	37,440
$IMUS_i^{\text{US} \rightarrow \text{CN}}$	0.113	0.082	0.002	0.036	0.182	0.226	37,440
$IMCN_i^{\text{US} \rightarrow \text{CN}}$	0.026	0.043	0.000	0.000	0.032	0.078	37,440
$IMUS_i^{\text{CN} \rightarrow \text{US}}$	0.089	0.070	0.002	0.027	0.140	0.188	37,440

Notes: See Section 3.1 and Appendix Section B.2 for the definitions of these variables.

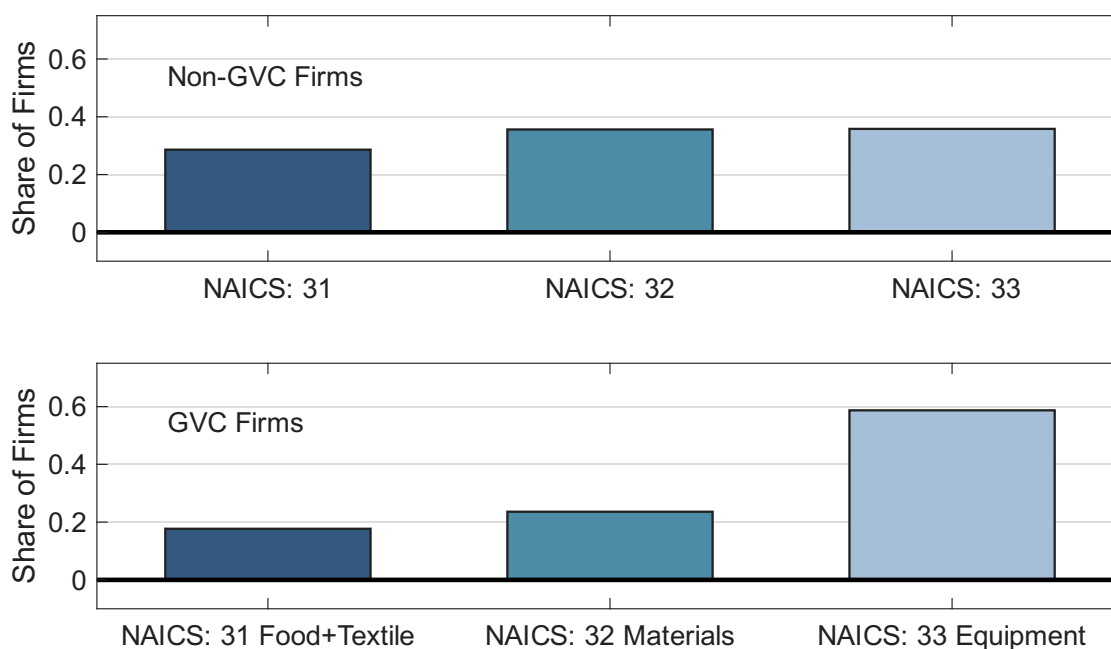


Figure A.1: Distribution of firms across industries

Notes: Top figure: Non GVC firms across two-digit industries (N=14,820). Bottom figure: GVC firms across two-digit industries (N=37,440)

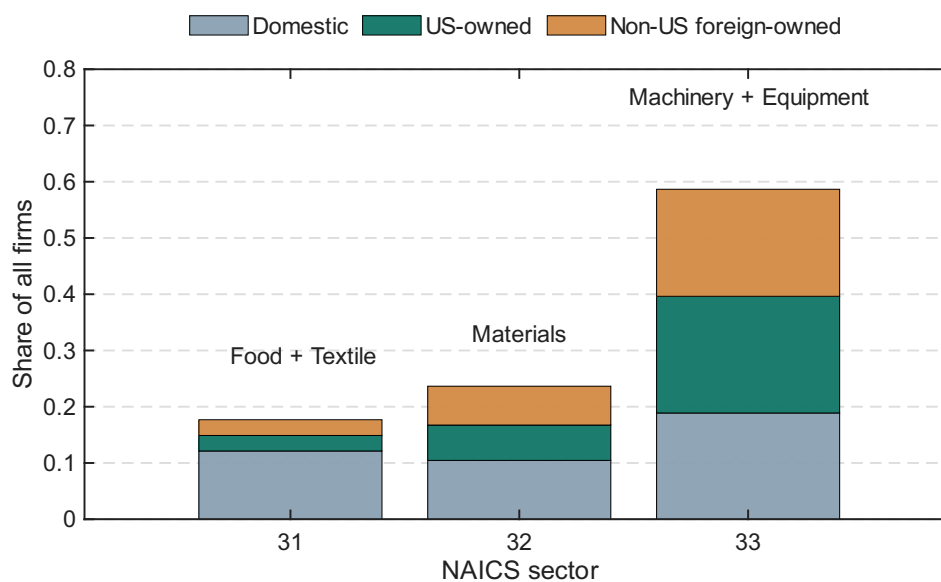


Figure A.2: Distribution of GVC firms across industries by ownership type

Notes: GVC firms by ownership type across two-digit industries (N=37,440)

Table A.2: Summary Statistics—GVC Firms by Country of Ownership

	(1)	(2)	(3)	(4)	(5)
Variable	Mean	SD	p25	p75	Obs
Panel. A U.S. Firms					
Log Employment	5.74	1.46	4.79	6.71	1183
Log Value of Production	16.91	1.68	15.91	18.02	1177
Log Value of Exports	16.24	2.42	15.03	17.96	1183
Log Value of Import	16.27	2.22	15.05	17.79	1170
US Export Share	0.89	0.23	0.92	1.00	1183
US Import Share	0.64	0.28	0.46	0.88	1170
Chinese Import Share	0.12	0.18	0.01	0.17	1170
Panel. B Non U.S. Foreign Firms					
Log Employment	5.69	1.51	4.69	6.72	1164
Log Value of Production	17.12	1.93	16.02	18.41	1157
Log Value of Exports	15.79	2.97	13.94	17.85	1164
Log Value of Import	16.38	2.27	15.19	17.84	1152
US Export Share	0.76	0.33	0.64	1.00	1164
US Import Share	0.39	0.32	0.10	0.67	1152
Chinese Import Share	0.14	0.20	0.01	0.19	1152
Panel. C Domestic Firms					
Log Employment	4.80	1.48	3.76	5.79	1745
Log Value of Production	15.72	1.92	14.48	16.93	1729
Log Value of Exports	14.34	2.44	12.94	15.95	1745
Log Value of Import	14.05	2.44	12.52	15.74	1647
US Export Share	0.83	0.31	0.82	1.00	1745
US Import Share	0.55	0.36	0.19	0.89	1647
Chinese Import Share	0.14	0.23	0.00	0.17	1647

Notes: GVC (IMMEX) firms in 2017 by country of ownership. Values are expressed in 2019 constant USD. Employment is measured in number of workers. Domestic firms are defined as those whose parent company is identified as domestic in the D&B database, or as firms that do not have a listed parent company.

Table A.3: Summary Statistics—IMMEX establishment-level data (2015-2023)

Variable	(1) Mean	(2) SD	(3) p25	(4) p75	(5) Obs
Log Total Revenue	16.19	1.88	14.98	17.46	37,822
Log Total Foreign Revenue	15.66	1.91	14.47	16.96	35,762
Log Sales of Own Manufactured Goods	16.22	2.73	15.16	17.99	20,766
Log Domestic Sales of Own Manufactured Goods	15.36	2.83	13.96	17.32	19,139
Log Foreign Sales of Own Manufactured Goods	15.73	2.32	14.51	17.26	17,234
Labor Cost Share	0.21	0.16	0.10	0.27	37,904
Material Cost Share	0.62	0.22	0.50	0.79	37,904
Foreign Material Cost Share	0.42	0.31	0.10	0.70	37,904
Log Non Production Employees	3.60	1.50	2.56	4.66	37,330
Log Production Employees	5.39	1.45	4.43	6.40	37,793
Log Employment on Firm's Payroll	5.48	1.55	4.53	6.55	33,342
Log Employment not on Firm's Payroll	4.71	1.82	3.58	6.00	10,465

Notes: IMMEX plants matched with PEME. Values are expressed in 2019 constant USD. Total revenue includes sales of own manufactured goods and revenues from contract manufacturing. The labor cost share is defined as total labor payments—including wages, benefits, and payments to staffing companies—divided by total expenses. The material cost share is total material costs divided by total expenses, and the foreign material cost share is total foreign material costs divided by total expenses. Non-production employees include administrative, managerial, and other non-production workers, whether directly hired or supplied through staffing companies. Production employees include production workers and technicians involved in manufacturing, whether on the firm's payroll or supplied through staffing companies.

B Data and Variable Construction

Firm-level data The Profile of Export-Oriented Manufacturing Firms (PEME) is a statistical product developed by Mexico's National Institute of Statistics and Geography (INEGI) that provides detailed firm-level information on the foreign trade activities of manufacturing firms. PEME is constructed by linking multiple data sources, including administrative foreign trade records, manufacturing surveys, and statistics from the IMMEX program. These datasets are integrated through Mexico's Statistical Business Register (RENEM), which provides a unified framework for identi-

fyng and tracking firms across sources. INEGI has undertaken this data-linking effort since 2013, combining foreign trade records with information from the Economic Censuses, the Annual and Monthly Manufacturing Industry Surveys, and IMMEX Program Statistics.

The unit of observation in PEME is the firm, defined as an organization owned by a single legal entity that engages in one or more economic activities. Firms are autonomous in decisions related to marketing, finance, and investment, and allocate resources according to a defined production strategy. A firm may operate in one or multiple establishments under the same legal name. PEME includes manufacturing firms for which at least one establishment reports information in the Manufacturing Surveys or the IMMEX Program Statistics.

Plant-level data The IMMEX plant-level dataset covers all establishments belonging to firms participating in the IMMEX Program, making it a census of activity conducted under the program. Since the analysis focuses on manufacturing, I restrict attention to manufacturing plants. The data, originally reported at the monthly frequency, are aggregated to the annual level.

Registry data In addition to the IMMEX plant-level data, I use the monthly IMMEX firm registry maintained by Mexico’s Ministry of Economy (Secretaría de Economía) for the period 2016–2024. The registry includes firm names, descriptions of primary activities (which roughly correspond to three-digit NAICS industries), addresses, and tax identifiers (Registro Federal de Contribuyentes, RFC). The registry is updated monthly to reflect plant openings and closures, which I use to identify new plant establishments. Firm tax identifiers are used to link firms to their global ultimate owners using D&B’s Corporate Hierarchies and Linkages database.

B.1 Overview of Categories, Variables, and Classifications

In the PEME, firms’ economic activity is classified using the North American Industry Classification System (NAICS, 2018), based on establishments’ outputs and production processes. Each firm is assigned a primary activity at the three-digit NAICS level, yielding 21 manufacturing industries. Importantly, the PEME includes microenterprises (1–10 employees), providing more

complete coverage of firms' contributions to employment and Mexico's international trade.

INEGI provides a crosswalk between establishment and firm identifiers for the period 2013–2023.

This key makes it possible to track the number of establishments belonging to a firm over time. It also enables the merge of firm-level PEME data with the plant-level IMMEX database.

Product classification as consumption goods is based on UNCTAD-SoP3 mapping provided at the World Integrated Trade Solution of the World Bank (<https://wits.worldbank.org/referencedata.html>).

B.2 Firm-Level Controls for Concurrent Trade Policy Changes

Section 232 and Mexico's retaliation Before the US government targeted China specifically, Section 232 tariffs led to retaliation from Mexico, among other countries. To control for the potentially confounding effects of the brief US-Mexico tariff escalation, we separately construct firm-specific exposures to the US tariffs on Mexican products ($T_i^{\text{US} \rightarrow \text{MEX}}$) and Mexico's retaliatory tariffs on US products ($T_i^{\text{MEX} \rightarrow \text{US}}$).

To construct firm-level exposure to the increased US tariffs on Mexico, we focus on firms that, before the shock, were exporting goods to the US that will be subject to Section 232 tariffs:

$$T_i^{\text{US} \rightarrow \text{MEX}} = \frac{\sum_{\ell \in \text{USIT}^{\text{MEX}}} X_{i\ell}^{2017, \text{US}} \times \Delta \tau_{\ell}^{\text{USIT}^{\text{MEX}}}}{\sum_{\ell} X_{i\ell}^{2017}} \quad (\text{B.1})$$

Since the Section 232 tariffs included other third countries, $T_i^{\text{US} \rightarrow \text{MEX}}$ does not cleanly identify the specific effects of US tariffs on Mexico but essentially functions as a robust control for the possible confounding effects of the Section 232 tariffs.

To measure the effect of the Mexico's retaliatory tariffs, we focus on firms that, before the shock, were importing goods from the US, which will later be subject to Mexico's retaliatory tariffs. Let USRT^{MEX} be the set of HS-6 products subject to the retaliatory tariffs of Mexico, and $\Delta \tau_j^{\text{USRT}^{\text{MEX}}}$ measures the change in the Mexican import tariff rate for good ℓ from the US in decimal points and let $M_{i\ell}^{2017}$ be the value of firm i 's imports of good ℓ in 2017, and let $M_{i\ell}^{2017, \text{US}}$ denote firm i 's imports of good ℓ from the US. The firm-level exposure to the Mexican government's retaliatory

tariffs is then given by the following:

$$T_i^{\text{MEX} \rightarrow \text{US}} = \frac{\sum_{\ell \in \text{USRT}^{\text{MEX}}} M_{i\ell}^{2017, \text{US}} \times \Delta \tau_{\ell}^{\text{USRT}^{\text{MEX}}}}{\sum_{\ell} M_{i\ell}^{2017}} \quad (\text{B.2})$$

As the empirical analysis focuses on exporters as of 2017, not all firms are importing in the sample. For non-importing firms, $T_i^{\text{MEX} \rightarrow \text{US}}$ is defined as zero.

Tariff Exposures via Input Channels I additionally construct firm-level exposure measures for firms that import inputs from the US exposed to the US tariffs on China, as well as for firms whose inputs concentrate on Chinese imports that would have been subject to US tariffs if imported into the US as follows:

$$IMUS_i^{\text{US} \rightarrow \text{CN}} = \frac{\sum_{\ell \in \text{USIT}^{\text{CN}}} M_{i\ell}^{2017, \text{US}} \times \Delta \tau_{\ell}^{\text{USIT}^{\text{CN}}}}{\sum_{\ell} M_{i\ell}^{2017}} \quad (\text{B.3})$$

$$IMCN_i^{\text{US} \rightarrow \text{CN}} = \frac{\sum_{\ell \in \text{USIT}^{\text{CN}}} M_{i\ell}^{2017, \text{CN}} \times \Delta \tau_{\ell}^{\text{USIT}^{\text{CN}}}}{\sum_{\ell} M_{i\ell}^{2017}} \quad (\text{B.4})$$

China's retaliatory tariffs By offering manufacturing and export services like polishing, packaging, and warehousing to US businesses, some exporting firms likely complement US operations without altering the nature of the goods traded, and hence their exports may be dampened due to costlier entry of US goods to China. However, particularly in the agricultural sector, Mexico's exports may also compete with US products in global markets.

Let USRT^{CN} represent the set of HS 6-digit products targeted by China's retaliatory tariffs, and $\Delta \tau_{\ell}^{\text{USRT}^{\text{CN}}}$ denote the increase in the retaliatory tariff rate for good ℓ as of December 2019.

$$T_i^{\text{CN} \rightarrow \text{US}} = \frac{\sum_{\ell \in \text{USRT}^{\text{CN}}} X_{i\ell}^{2017} \times \Delta \tau_{\ell}^{\text{USRT}^{\text{CN}}}}{\sum_i X_{i\ell}^{2017}} \quad (\text{B.5})$$

$T_i^{\text{CN} \rightarrow \text{US}}$ measures the weighted value of exports in goods that are targeted by China's retaliatory tariffs relative to the total exports of firm i as of 2017. As before, we weight exported goods based

on their respective retaliatory tariff rates, ensuring that Mexican firms with a greater concentration of exports in goods subject to higher export tariffs are regarded as more exposed, all else being equal.

I additionally account for firms whose imports from the US concentrated in goods that will be subject to retaliatory tariffs from China.

$$IMUS_i^{CN \rightarrow US} = \frac{\sum_{\ell \in USRT^{CN}} M_{i\ell}^{2017, US} \times \Delta \tau_{\ell}^{USRT^{CN}}}{\sum_{\ell} M_{i\ell}^{2017}} \quad (B.6)$$

Table A.1 presents summary statistics for these measures.

B.3 Sample Restriction for the U.S.-Specific Exposure Measure

The U.S.-specific exposure measure is designed to capture variation among U.S.-exporting firms based on the extent to which their sales are concentrated in goods subject to U.S. tariffs. When firms have low baseline U.S. export shares, this measure becomes less informative about their exposure to U.S. tariffs, as their U.S.-oriented sales constitute only a small portion of overall activity. Table B.1 examines how firms' U.S. export shares relate to tariff exposure across subsamples defined by baseline U.S. export intensity. The estimated relationship is driven by firms with low initial U.S. export shares and declines as the minimum U.S. export-share threshold is raised and becomes statistically insignificant at around the 30 percent cutoff, which motivates the baseline sample restriction.

Table B.1: U.S. Export Share Gains by Baseline U.S. Export Share

Sample based on 2017 share of US exports		(1) All ($>=0$)	(2) >0	(3) $>=0.1$	(4) $>=0.2$	(5) $>=0.3$	(6) $>=0.4$
Dep var. for all columns		US Export Share					
$T_i^{\text{US} \rightarrow \text{CN}} \times \text{PostTW}_t$		0.120*** (0.039)	0.085** (0.039)	0.071* (0.041)	0.062 (0.041)	0.053 (0.040)	0.027 (0.039)
N		50,550	46,596	43,614	42,335	41,265	40,054
Firm FE		✓	✓	✓	✓	✓	✓
3-digit Industry $\times$ Year FEs		✓	✓	✓	✓	✓	✓
Additional Trade Policy Controls		✓	✓	✓	✓	✓	✓

Notes: PostTW_t is a dummy variable equal to 1 for years 2018 and later, and 0 otherwise. Additional Trade Policy Controls include firms' exposures to the US-Mexico tariff escalation, retaliatory tariffs of China and the input channels as described in equations B.1-B.6. Robust standard errors are clustered at the firm level.

C Additional Analysis

C.1 Firms' Trade Outcomes

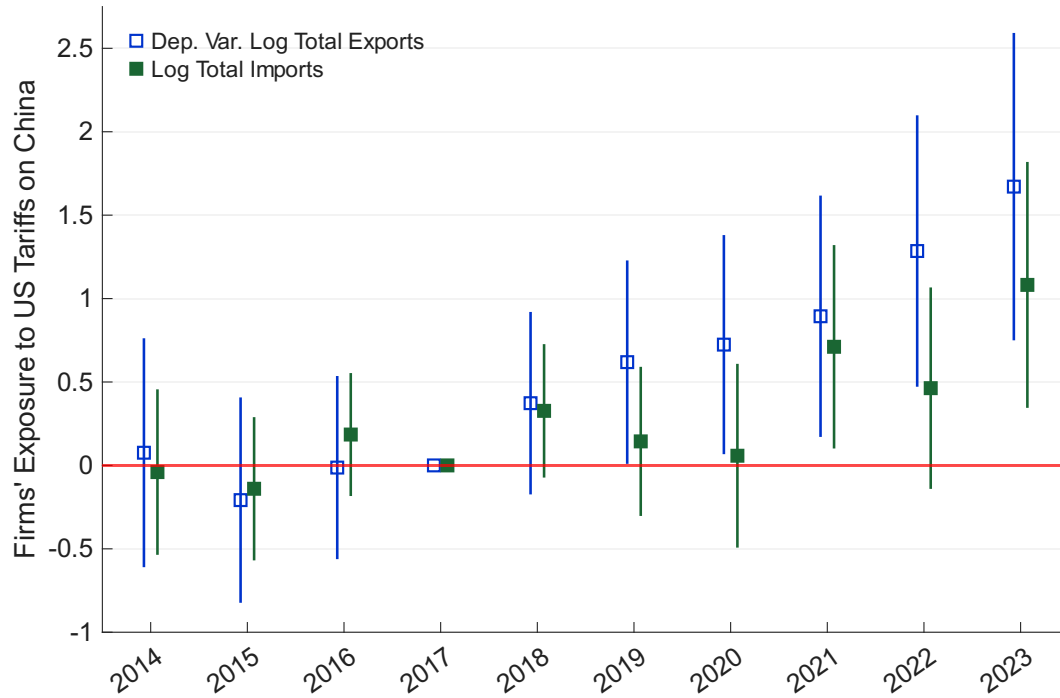


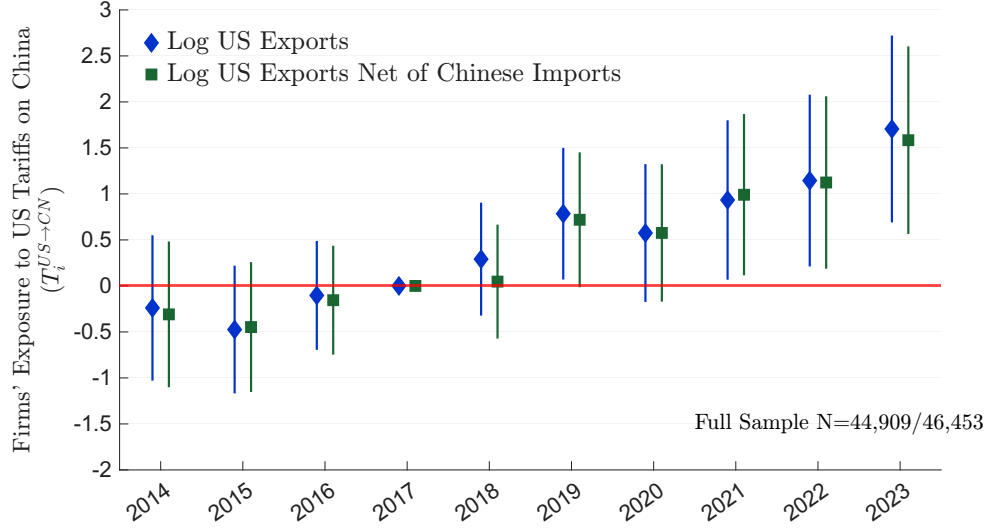
Figure C.1: Impact of US Tariffs Targeting China on Manufacturing Firms' Total Exports and Imports

Notes: Estimation of equation 3. The dependent variables are the natural logarithm of firms' value of exports (N=50,550) and imports (49,706). All regressions control for firm fixed effects, three-digit-industry-by-year fixed effects and concurrent trade policy changes. The y-axis displays the yearly DD coefficient estimates corresponding to $T_i^{US \rightarrow CN}$, along with 95% confidence intervals based on robust standard errors clustered at the firm level.

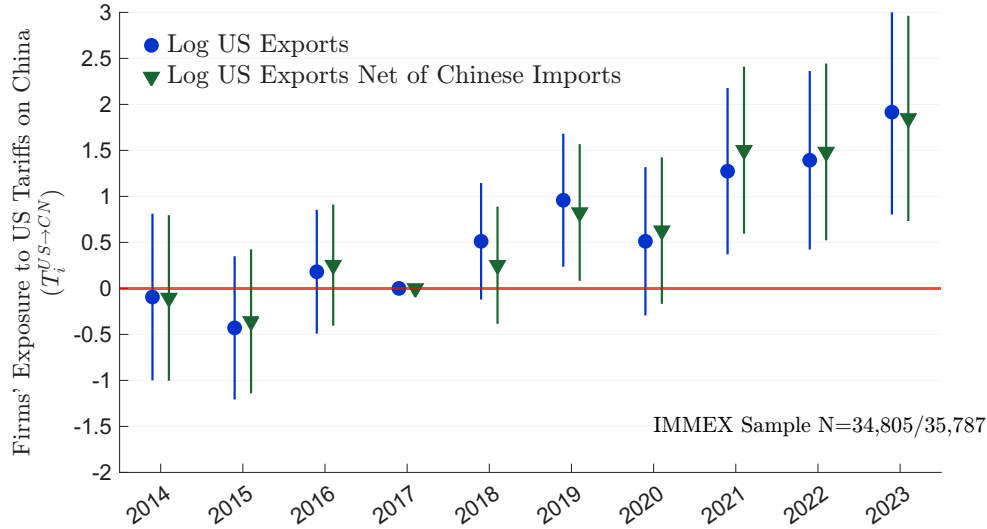
Table C.1: The US-China Trade War and Manufacturing Firms' Trade Outcomes

Panel A. IMMEX-GVC Firms							
	(1) Log US Exports	(2) Log Total Exports	(3) Log Non-US Exports	(4) Log Total Imports	(5) Log US Imports	(6) Log Non-US Imports	(7) Share of Imports from China
$T_i^{US \rightarrow CN} \times PostTW_t$	1.157*** (0.331)	1.088*** (0.302)	-0.615 (0.503)	0.585** (0.256)	0.604* (0.317)	0.736** (0.300)	0.019 (0.037)
N	35,787	36,859	22,381	36,430	35,217	35,219	36,430
Panel B. Other firms							
$T_i^{US \rightarrow CN} \times PostTW_t$	0.730 (0.757)	0.556 (0.547)	1.012 (0.633)	0.407 (0.411)	0.521 (0.526)	0.038 (0.493)	0.052 (0.050)
N	10,666	13,691	10,779	13,276	11,716	12,729	13,276
For both panels:							
Firm FEs	✓	✓	✓	✓	✓	✓	✓
3-digit Industry $\times$ Year FEs	✓	✓	✓	✓	✓	✓	✓
Additional Trade Policy Controls	✓	✓	✓	✓	✓	✓	✓

Notes: $PostTW_t$ is a dummy variable equal to 1 for years 2018 and later, and 0 otherwise. Additional Trade Policy Controls include firms' exposures to the US-Mexico tariff escalation, retaliatory tariffs of China and the input channels as described in equations B.1-B.6. Robust standard errors are clustered at the firm level.



(a) Full Sample



(b) GVC Firms

Figure C.2: Impact of US Tariffs Targeting China on US Exports Net of Chinese Imports

Notes: Estimation of equation 3. The dependent variables are the natural logarithm of US exports and US exports net of Chinese imports, $\sum_{\ell \in \mathcal{L}_{it}^{US}} [X_{it}^{US} - M_{it}^{CN}]$. All regressions control for firm fixed effects, three-digit-industry-by-year fixed effects and concurrent trade policy changes. The y-axis displays the yearly DD coefficient estimates corresponding to $T_1^{US \rightarrow CN}$, along with 95% confidence intervals based on robust standard errors clustered at the firm level.

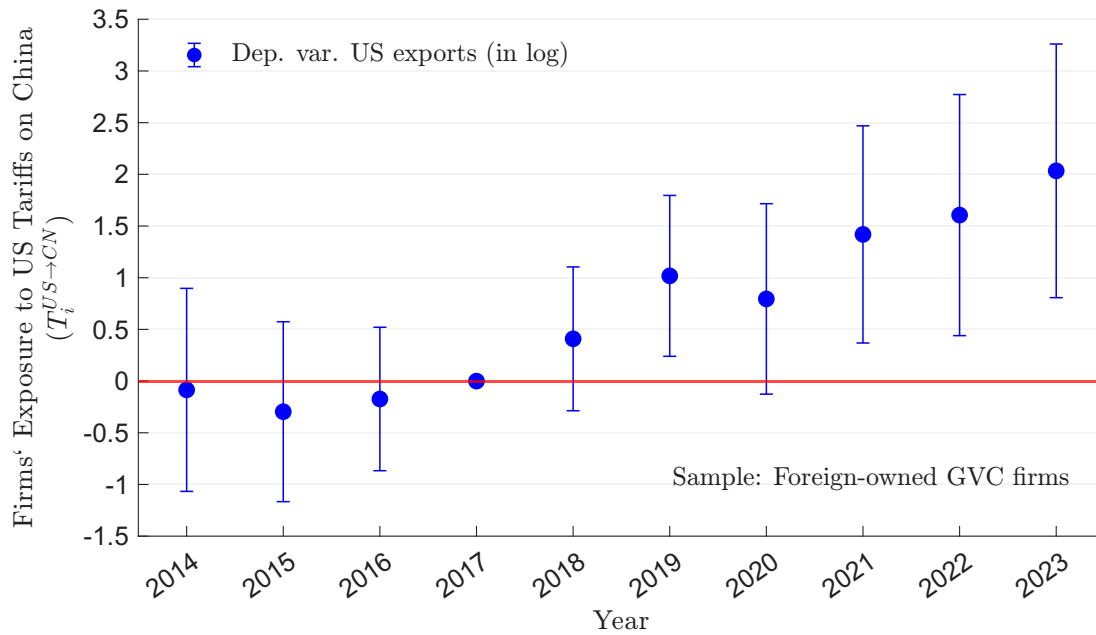


Figure C.3: How US Exports of Foreign GVC Firms Respond to the US Tariffs

Notes: The dependent variables are the natural logarithm of US exports, N=21,177 (among foreign IMMEX). All regressions control for firm fixed effects, industry-by-year fixed effects and concurrent trade policy changes. The y-axis displays the yearly DD coefficient estimates corresponding to $T_i^{US \rightarrow CN}$, along with 95% confidence intervals based on robust standard errors clustered at the firm level.

C.2 Supplemental Analysis—Manufacturing Output

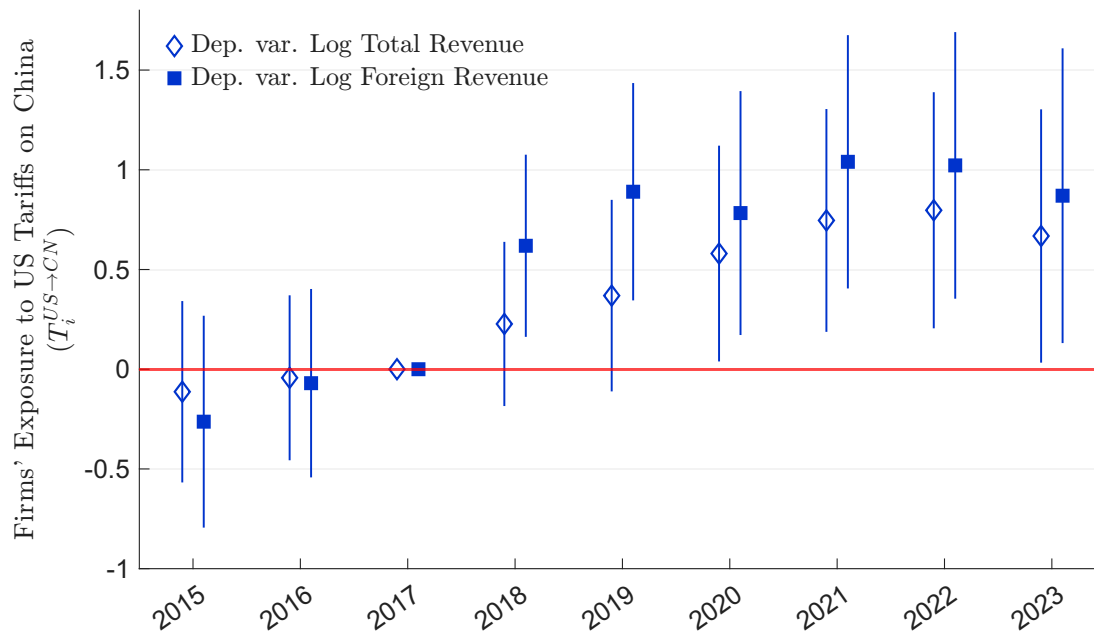


Figure C.4: Impact of the U.S. Tariff Hikes Targeting China on Plants' Revenue

Notes: Estimation of equation 4. The dependent variables are the natural logarithms of total plant revenue (N=36,378) and total foreign revenue (N=34,325). All regressions control for plant fixed effects, three-digit-industry-by-year, municipality-by-year fixed effects and concurrent trade policy changes. The y-axis displays the yearly DD coefficient estimates corresponding to $T_i^{US \rightarrow CN}$, along with 95% confidence intervals based on robust standard errors clustered at the firm level.

Table C.2: Impact of U.S. Tariffs Targeting China on Plant-Level Revenue and Sales

Panel A: Total Revenue and Sales			
	(1) Log Total Revenue	(2) Log Total Sales	(3) Log Contract Manufacturing Revenue
$\mathbf{T}_i^{\text{US} \rightarrow \text{CN}} \times \text{PostTW}_t$	0.548*** (0.193)	1.209*** (0.368)	0.236 (0.304)
Observations	36,378	19,410	19,996
R-squared	0.939	0.953	0.925
Panel B: Foreign Revenue and Sales			
	(4) Log Total Foreign Revenue	(5) Log Foreign Sales	(6) Log Foreign Contract Manufacturing Revenue
$\mathbf{T}_i^{\text{US} \rightarrow \text{CN}} \times \text{PostTW}_t$	0.959*** (0.245)	1.729*** (0.441)	0.349 (0.274)
Observations	34,325	15,967	18,759
R-squared	0.909	0.911	0.923
For both panels:			
Plant Fixed Effects	✓	✓	✓
Industry (3-digit) $\times$ Year FE	✓	✓	✓
Municipality $\times$ Year FE	✓	✓	✓

Notes: The table reports estimates of Equation 4 using IMMEX plant-level data. The dependent variables are the natural logarithms of revenue and sales measures as indicated in the column headers. Contract manufacturing revenue refers to revenues from production services provided to other firms (maquila). All regressions include plant fixed effects, three-digit industry-by-year fixed effects, municipality-by-year fixed effects and controls for concurrent trade policy changes. Robust standard errors clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table C.3: Impact of U.S. Tariffs Targeting China on Material Inputs and Value Added

	(1) Log Foreign Material Expenses	(2) Log Domestic Material Exp.	(3) Log Total Material Exp.	(4) Log Value Added	(5) Log Domestic VA in Exports
$T_i^{US \rightarrow CN} \times \text{PostTW}_t$	1.476*** (0.392)	0.249 (0.435)	0.595* (0.305)	0.420** (0.209)	0.610** (0.252)
Observations	30,800	19,766	36,269	33,313	31,173
R-squared	0.890	0.898	0.897	0.910	0.900
For all specifications:					
Plant Fixed Effects	✓	✓	✓	✓	✓
Industry (3-digit) $\times$ Year FE	✓	✓	✓	✓	✓
Municipality $\times$ Year FE	✓	✓	✓	✓	✓

Notes: The table reports estimates of Equation 4 using IMMEX plant-level data. The dependent variables are the natural logarithms of material expenditures and value added measures.

Value added is defined as total sales minus material and packaging input expenditures, plus revenue from contract manufacturing services (maquila), which captures payments for processing activities not included in sales.

Domestic value added in exports is defined as foreign sales minus foreign material and packaging input expenditures, plus revenue from foreign contract manufacturing. This construction assumes that all foreign inputs are used in export-oriented production. If some foreign inputs are instead used for domestic sales, the measure will understate domestic value added embodied in exports.

All regressions include plant fixed effects, three-digit industry-by-year fixed effects, municipality-by-year fixed effects, and controls for concurrent trade policy changes. Robust standard errors clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.3 Supplemental Analysis—Manufacturing Employment

Table C.4: The US-China Trade War and Manufacturing Employment

Dep var. Log Employment	Broad Exposure			US market specific		
	ALL	GVCs	Others	ALL	GVCs	Others
2014	-0.057 (0.088)	-0.117 (0.134)	-0.007 (0.099)	0.099 (0.116)	0.069 (0.156)	0.195 (0.145)
2015	0.017 (0.081)	-0.051 (0.110)	0.095 (0.122)	0.120 (0.098)	0.084 (0.131)	0.231* (0.120)
2016	0.015 (0.061)	-0.030 (0.073)	0.080 (0.107)	0.064 (0.066)	0.057 (0.088)	0.110 (0.078)
2018	0.067 (0.100)	0.084 (0.116)	-0.022 (0.190)	0.127 (0.103)	0.163 (0.126)	-0.018 (0.163)
2019	0.121 (0.109)	0.132 (0.129)	0.060 (0.205)	0.115 (0.118)	0.125 (0.141)	0.039 (0.202)
2020	0.176 (0.116)	0.167 (0.132)	0.171 (0.231)	0.277** (0.129)	0.197 (0.147)	0.490* (0.261)
2021	0.275** (0.128)	0.359** (0.164)	0.157 (0.192)	0.382*** (0.146)	0.452*** (0.175)	0.157 (0.243)
2022	0.249* (0.138)	0.351** (0.169)	0.081 (0.227)	0.425*** (0.154)	0.445** (0.182)	0.377 (0.276)
2023	0.374** (0.145)	0.518*** (0.183)	0.118 (0.224)	0.508*** (0.162)	0.570*** (0.194)	0.269 (0.276)
N	52,174	37,385	14,789	42,171	33,573	8,598
Firm FEs	✓	✓	✓	✓	✓	✓
Industry × Year FEs	✓	✓	✓	✓	✓	✓
Concurrent Trade Policy Controls	✓	✓	✓	✓	✓	✓

Notes: Concurrent Trade Policy Controls include firms' exposures to the US-Mexico tariff escalation, retaliatory tariffs of China and the input channels as described in equations B.1-B.6. Robust standard errors are clustered at the firm level.

C.4 Aggregation Across Establishments and Single-Plant Firms

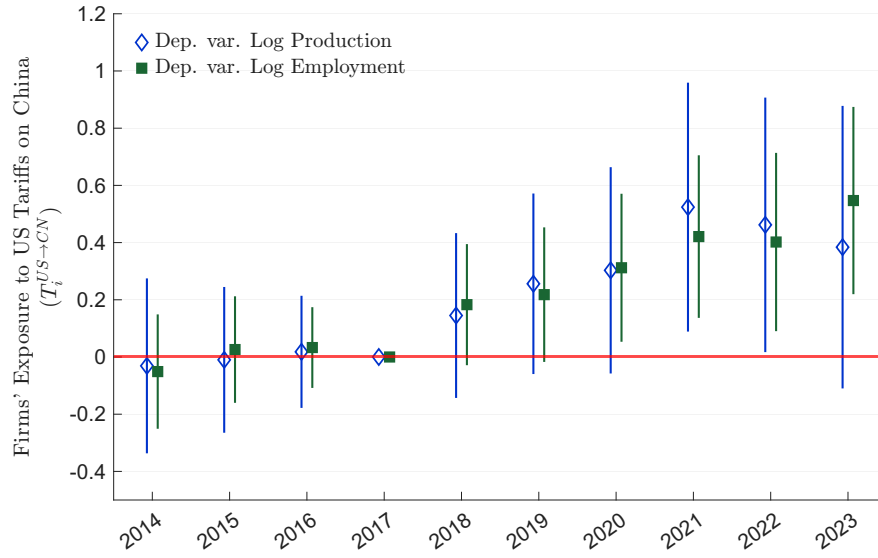
Firm-level measures of production and employment in the PEME database are constructed by aggregating information across establishments whose primary activity is classified as manufacturing. This aggregation can attenuate estimated responses if firms adjust along margins that are not fully captured within the set of establishments included in the data. In particular, firms may reallocate production or employment across establishments or expand activity through the addition of new plants that are not immediately reflected in firm-level measures. In such cases, firm-level estimates may understate the true extent of adjustment.

To assess the importance of this concern, I estimate Equation 3 on a subsample of firms that operated a single manufacturing establishment in 2017. Focusing on this group limits the scope for reallocation across pre-existing plants and provides a cleaner measure of within-establishment adjustment.

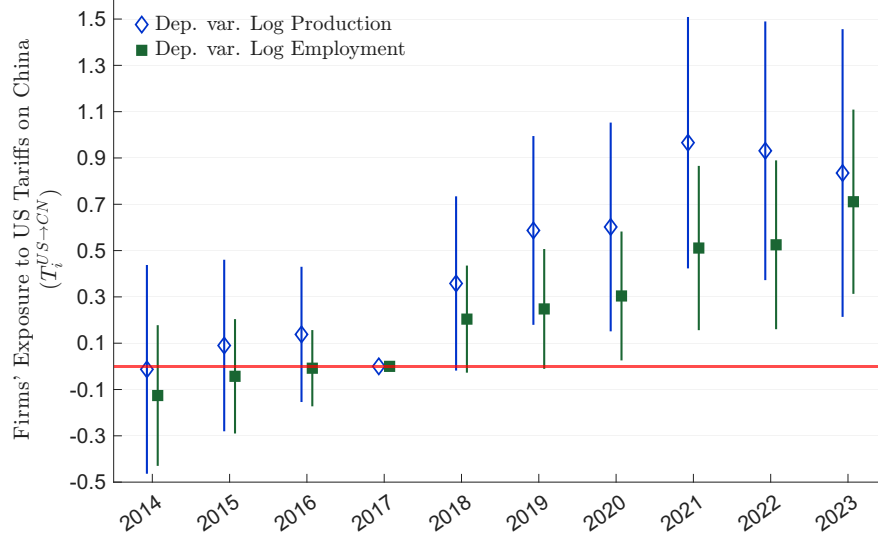
Figure C.5 reports the results. Panel (a) presents estimates for all single-plant exporters, while Panel (b) focuses on single-plant GVC firms. The results show a significant positive effect of U.S. tariffs on both production and employment. By 2023, the employment effect reaches 0.55 for all exporters and 0.71 for GVC firms, while production effects peak earlier, in 2021, with DD estimates of 0.52 and 0.97, respectively.

Comparing production and employment dynamics reveals that output initially expands more rapidly than labor input, particularly among GVC firms. Because the analysis focuses on firms operating a single establishment in 2017, this pattern primarily reflects within-establishment adjustment rather than reallocation across pre-existing plants. After 2021, employment growth accelerates and the gap narrows, suggesting that early output expansion was partly supported by increases in labor productivity (see Figure C.6).

Overall, these results indicate that the baseline firm-level estimates are not driven by aggregation across establishments and, if anything, may understate the magnitude of firms' responses.



(a) Single-plant Exporters



(b) Single-plant GVC firms

Figure C.5: Employment and Production Effect among Single Plant Firms

Notes: Estimation of equation 3 among single plant firms as of 2017. In figure (a) $N=42,596/42,297$, in figure (b) $N=30,853/30,694$. All regressions control for firm fixed effects, three-digit industry-by-year fixed effects and concurrent trade policy changes. The y-axis displays the yearly DD coefficient estimates corresponding to $T_i^{US \rightarrow CN}$, along with 95% confidence intervals based on robust standard errors clustered at the firm level.

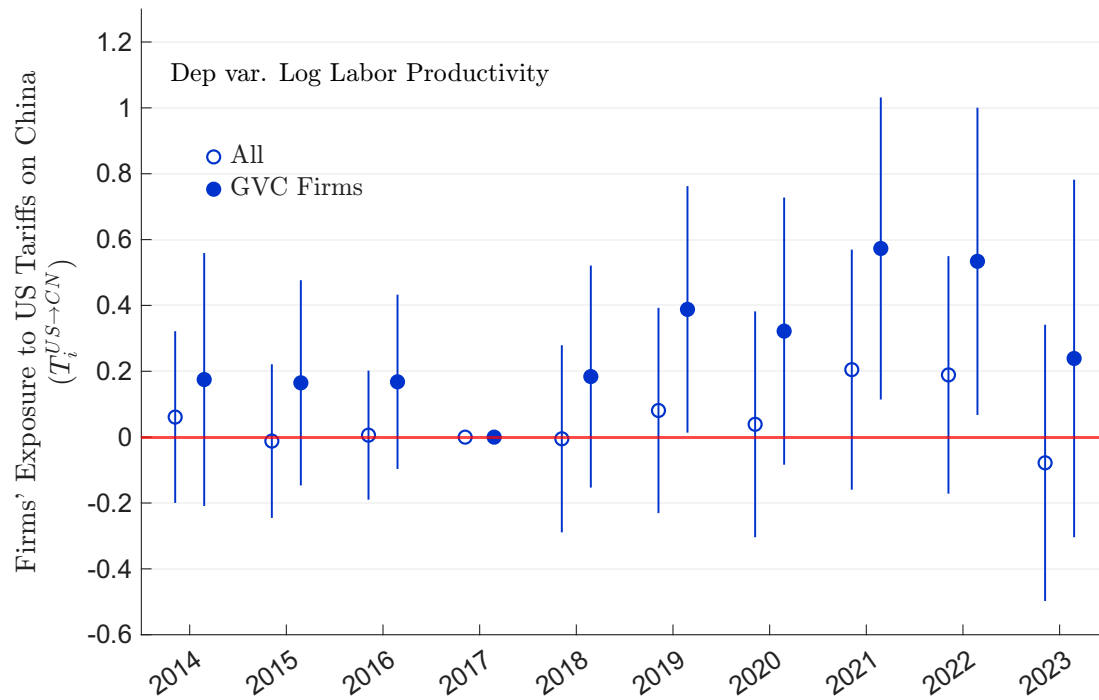


Figure C.6: The Effect of Rising U.S. Import Tariffs on China on Labor Productivity

Notes: Estimation of equation 3 among single-plant firms with the dependent variable being the natural logarithm of the labor productivity. $N=42,278/30,681$. All regressions control for firm fixed effects, three-digit industry-by-year fixed effects and concurrent trade policy changes. The y-axis displays the yearly DD coefficient estimates corresponding to $T_i^{US \rightarrow CN}$, along with 95% confidence intervals.

C.5 Within-Plant Adjustment: Additional Results

Table C.5: Impact of U.S. Tariffs Targeting China on Wages

	(1) Log Average Wage	(2) Log Average White-Collar Wage	(3) Log Average Blue-Collar Wage
$\mathbf{T}_i^{\text{US} \rightarrow \text{CN}} \times \text{PostTW}_t$	0.096 (0.101)	0.294 (0.180)	0.028 (0.099)
Observations	36,430	30,453	30,650
R-squared	0.805	0.778	0.788
Plant Fixed Effects	✓	✓	✓
Industry (3-digit) $\times$ Year FE	✓	✓	✓
Municipality $\times$ Year FE	✓	✓	✓

Notes: The table reports estimates of Equation 4 using IMMEX plant-level data. The dependent variables are the natural logarithms of average wages. Average wage is constructed by dividing total labor costs, inclusive of payments to non-payroll workers, by total employment. White-collar and blue-collar wages correspond to average hourly wages of payroll workers, excluding benefits, measured as of November. All regressions include plant fixed effects, three-digit industry-by-year fixed effects, municipality-by-year fixed effects, and controls for concurrent trade policy changes. Robust standard errors clustered at the firm level are reported in parentheses.

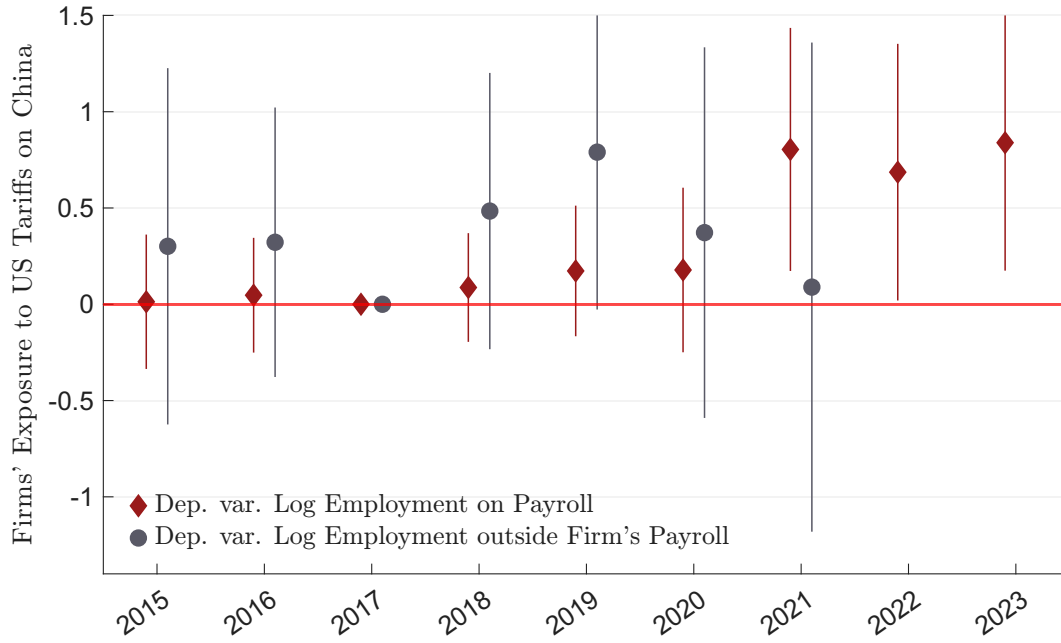


Figure C.7: Rising U.S. Tariffs on China and Labor Adjustment Margins

Notes: Estimation of equation 4 among IMMEX plants. The dependent variables are payroll employment ($N = 31,938$) and non-payroll employment ($N = 9,426$). All specifications include plant FE, three-digit-industry-by-year FE, municipality-by-year FE and concurrent trade policy changes. The y-axis displays the yearly DD coefficient estimates corresponding to $\mathbf{T}_i^{\text{US} \rightarrow \text{CN}}$, along with 95% confidence intervals. Robust SEs clustered at the firm level.

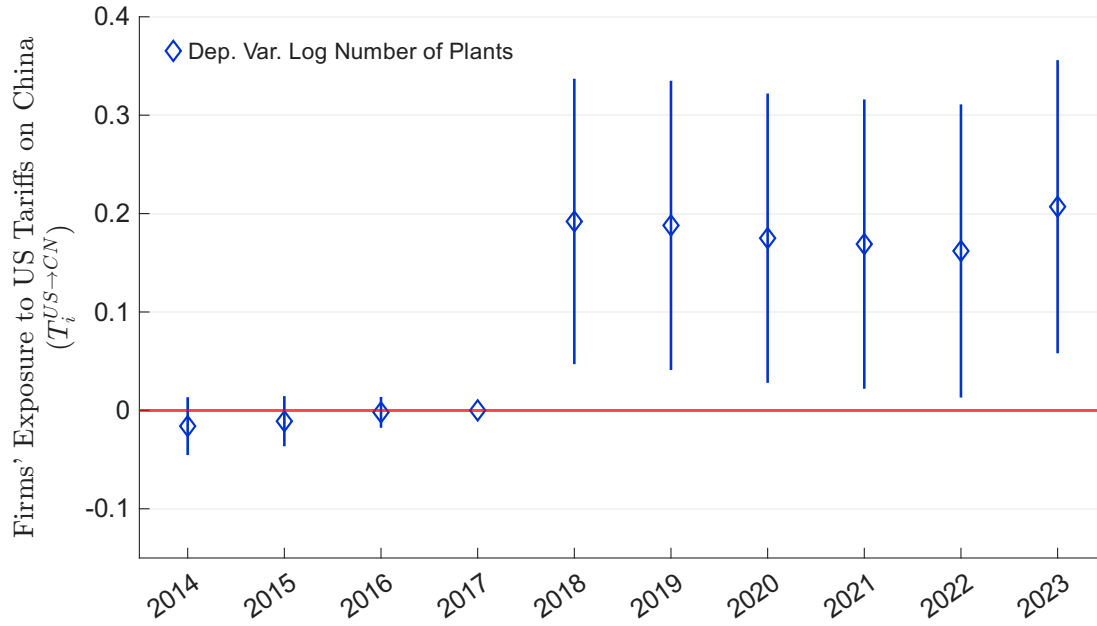


Figure C.8: Rising US import tariffs on China spur plant openings in Mexico

Notes: Estimation of Equation 3 with the dependent variable being the natural logarithm of number of plants a firm has. PEME sample firms exporting as of 2017, covering the period from 2014 to 2023, N=51,651. All regressions control for firm fixed effects, industry-by-year fixed effects and concurrent trade policy changes. The y-axis displays the yearly DD coefficient estimates corresponding to $T_i^{US \rightarrow CN}$, along with 95% confidence intervals.

C.6 Adjustment of U.S.-headquartered firms

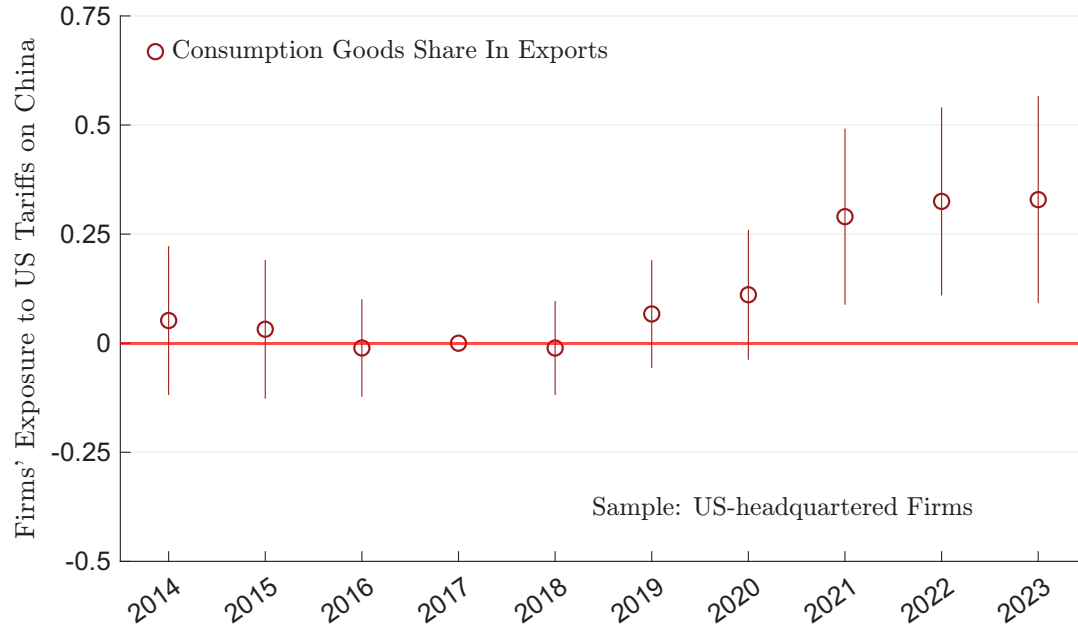


Figure C.9: US MNE Export Growth Tilts Toward Final Goods

Estimates of equation 3 for U.S.-headquartered ($N = 11,060$) firms. The dependent variable is the share of firm exports in consumption goods. All regressions include firm fixed effects, three-digit industry-by-year fixed effects, and controls for concurrent trade policy changes. The y-axis reports the yearly DD coefficient estimates corresponding to $T_1^{US \rightarrow CN}$, with 95% confidence intervals.

Table C.6: Export and Import Responses of US- and Non-US Foreign Firms Operating in Mexico

	(1) Log US Exports	(2) Log US Exports Net of Chinese Imports	(3) Log US Imports	(4) Log Chinese Imports
Panel A. US MNEs in Mexico				
$T_i^{\text{US} \rightarrow \text{CN}} \times \text{PostTW}_t$	1.000** (0.498)	0.902* (0.488)	1.074*** (0.380)	2.073*** (0.783)
N	10,940	10,706	10,955	10,459
Panel B. Non-US Foreign MNEs in Mexico				
$T_i^{\text{US} \rightarrow \text{CN}} \times \text{PostTW}_t$	1.548* (0.850)	1.539* (0.875)	0.715 (0.686)	0.398 (0.822)
N	10,231	9,809	10,369	9,954
For both panels				
Firm FE	✓	✓	✓	✓
Industry (3-dig) $\times$ Year FE	✓	✓	✓	✓
Concurrent Trade Policy Controls	✓	✓	✓	✓

Notes: PostTW_t is a dummy variable equal to 1 for years 2018 and later, and 0 otherwise. Additional Trade Policy Controls include firms' exposures to the US-Mexico tariff escalation, retaliatory tariffs of China and the input channels as described in equations B.1-B.6. Robust standard errors are clustered at the firm level.

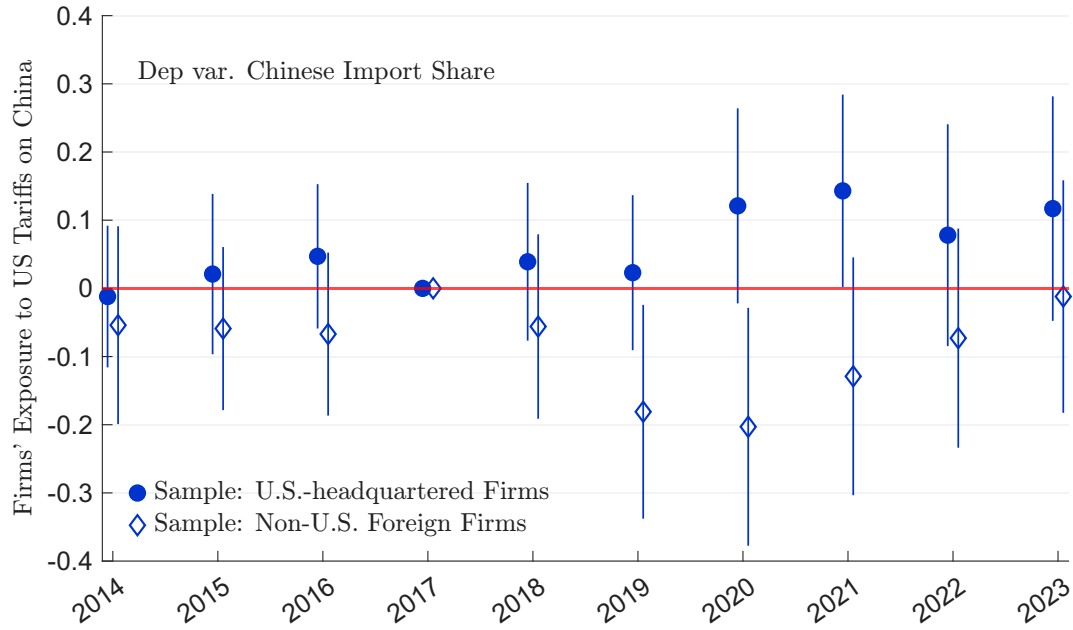


Figure C.10: Import Share from China in Response to Rising US Tariffs: U.S.- vs. Other Foreign-Headquartered Firms

Estimates of equation 3 for U.S.-headquartered ($N = 10,459$) and non-U.S. foreign-headquartered ($N = 9,954$) firms. The dependent variable is the share of firm imports sourced from China (imports from China divided by total imports). All regressions include firm fixed effects, three-digit industry-by-year fixed effects, and controls for concurrent trade policy changes. The y-axis reports the yearly DD coefficient estimates corresponding to $T_i^{US \rightarrow CN}$, with 95% confidence intervals.

D Robustness Analysis

This section examines the robustness of the main results to alternative specifications and potential confounding policy changes. In particular, it assesses the stability of the estimates to the sequential inclusion of control variables, alternative specifications for plant openings, and the introduction of USMCA-related policy changes.

D.1 Sequential Inclusion of Controls

This subsection examines the robustness of the main results to the sequential inclusion of control variables. Figures D.1–D.3 report estimates for U.S. exports, production, and employment, respec-

tively, under progressively richer specifications: firm and year fixed effects only; adding controls for concurrent trade policy measures; and further including two-digit and three-digit industry-by-year fixed effects.

Across all outcomes, the estimated effects remain stable in magnitude and statistical significance as controls are added.

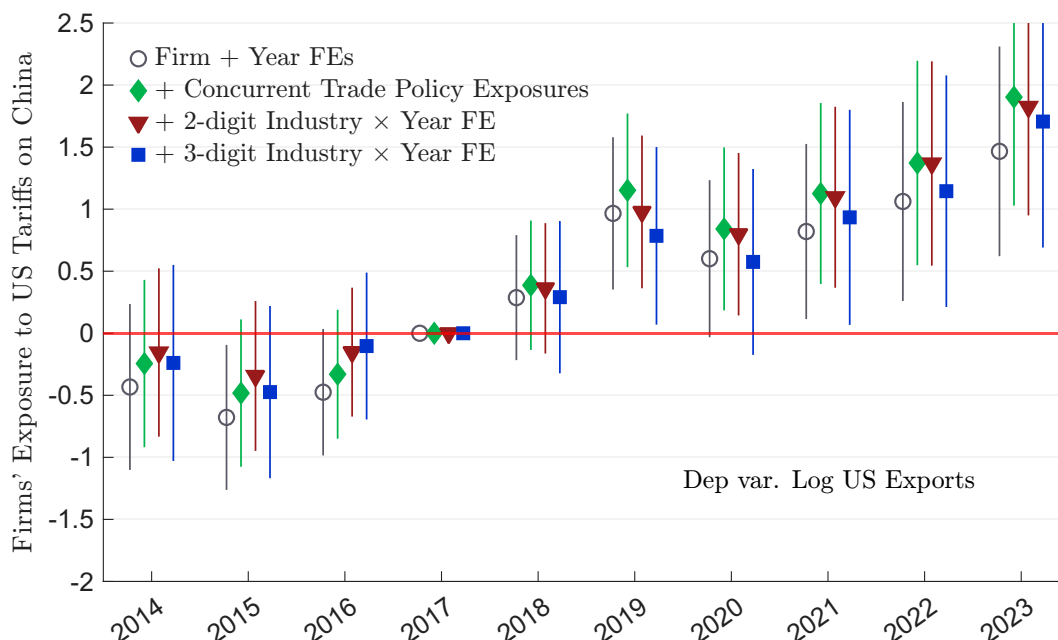


Figure D.1: U.S. tariffs on China and firms' exports to the United States, with sequentially added controls

Notes: Estimation of equation 3 with the dependent variable being the natural logarithm of firm's exports to the U.S.. $N=46,453$. The y-axis displays the yearly DD coefficient estimates corresponding to $T_i^{US \rightarrow CN}$, along with 95% confidence intervals.

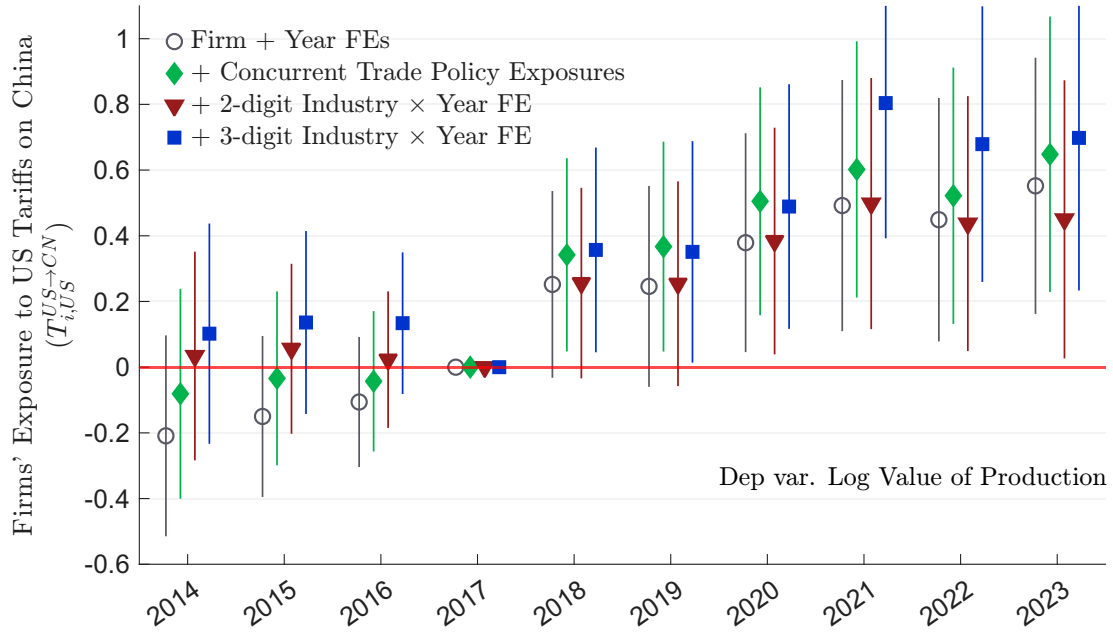


Figure D.2: U.S. tariffs on China and firms' production, with sequentially added controls

Notes: Estimation of equation 3 with the dependent variable being the natural logarithm of firm's value of production at factory prices. Estimation is conducted among U.S. exporters with $N=41,914$. The y-axis displays the yearly DD coefficient estimates corresponding to $T_{i,US}^{US \rightarrow CN}$, along with 95% confidence intervals.

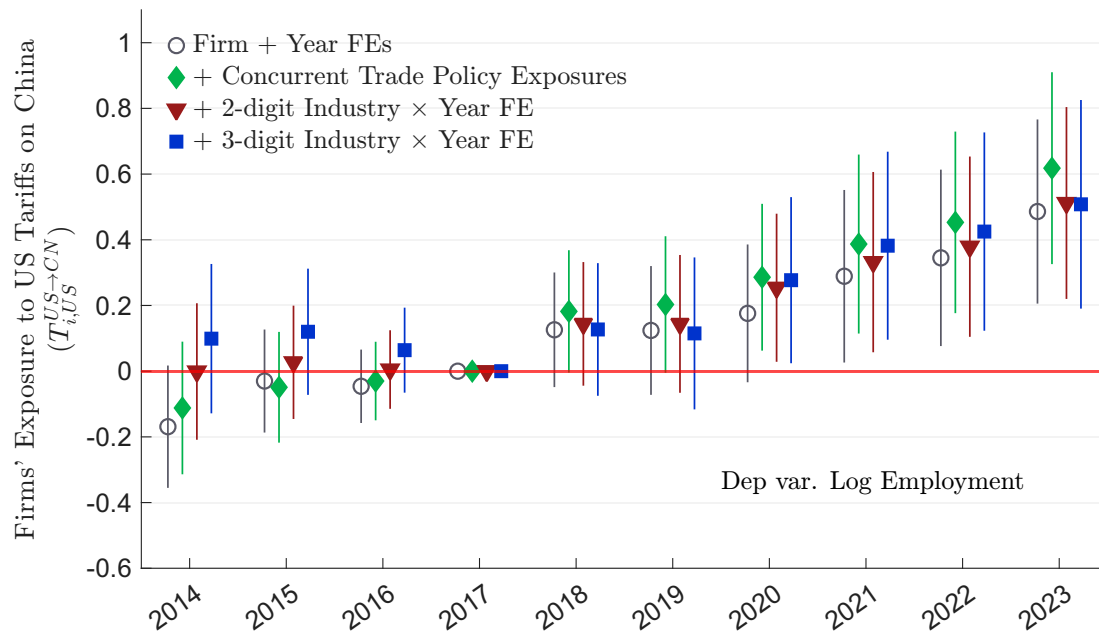


Figure D.3: U.S. tariffs on China and firms' employment, with sequentially added controls

Notes: Estimation of equation 3 with the dependent variable being the natural logarithm of firm's employment. Estimation is conducted among U.S. exporters with $N=42,171$. The y-axis displays the yearly DD coefficient estimates corresponding to $T_{i,US}^{US \rightarrow CN}$, along with 95% confidence intervals.

D.2 Robustness—Plant Openings in Mexico

This subsection examines the robustness of the plant-opening results to alternative specifications and inference procedures. Table [D.1](#) reports estimates using wild cluster bootstrap inference and continuous measures of Chinese input dependence, while Table [D.2](#) presents Poisson pseudo-maximum likelihood (PPML) estimates for the number of new plants. The results are broadly consistent with the baseline findings and confirm the role of Chinese input dependence and firm heterogeneity.

Across specifications, the results are consistent with the baseline findings.

Table D.1: Plant Openings in Mexico: Continuous Chinese Input Dependence and Wild Cluster Bootstrap Inference

	Linear			Linear $\times$ CN Input Dep.			
	(1) All	(2) Foreign	(3) Not Foreign	(4) All	(5) Foreign	(6) Not Foreign	(8) Foreign
$T_j^{US \rightarrow CN} \times PostTW_t$	0.590** (0.210)	0.791* (0.395)	0.317*** (0.099)	-2.086 (1.274)	-4.960*** (1.269)	0.450 (0.594)	0.723* (0.403)
$T_j^{US \rightarrow CN} \times PostTW_t \times IMCN_j^{US \rightarrow CN}$				111.974* (54.780)	228.110*** (55.939)	-4.693 (28.063)	334.898*** (62.432)
$T_j^{US \rightarrow CN} \times PostTW_t \times CN \text{ Affiliate}$							1.963*** (0.674)
Observations	33,797	17,193	16,604	33,627	17,166	16,461	17,193
R-squared	0.290	0.284	0.289	0.289	0.285	0.283	0.285
Bootstrap p: tariff exposure	0.016	0.068	0.159	0.308	0.030	0.571	0.104
Bootstrap p: interaction				0.166	0.021	0.902	0.698

Notes: Manufacturing IMMEX firms operating as of 2017, over 2017–2024. Dependent variable is the inverse hyperbolic sine transformation of the number of new plants. All specifications include firm and border $\times$ year fixed effects. Concurrent Trade Policy Controls include exposures to the U.S.–Mexico tariff escalation, China’s retaliatory tariffs, and the input channel measures $IMUS_j^{CN \rightarrow US}$, and $IMUS_j^{US \rightarrow CN}$, all constructed using firm-level data aggregated to the industry level and interacted with the post-trade war indicator. Columns (4)–(7) further control for Chinese Input Dependence, $IMCN_j^{US \rightarrow CN}$. CN Affiliate is a dummy variable that takes the value of one if the parent company headquarters are located in China, Hong Kong, or Singapore. All specifications include firm and border $\times$ year fixed effects. Robust standard errors are clustered at the three-digit industry level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Bootstrap p-values from wild cluster bootstrap (Roodman et al. 2019) reported at bottom.

Table D.2: Plant openings in Mexico: PPML Estimates

Dependent variable: Number of New Plants.						
	(1)	(2)	(3)	(4)	(5)	(6)
Sample:	All	All	Foreign	Not Foreign	US	Foreign
$\mathbf{T}_j^{\text{US} \rightarrow \text{CN}} \times \text{PostTW}_t$	4.631** (2.314)	0.010 (1.404)	-1.331 (1.867)	5.919*** (1.590)	-2.577 (1.642)	4.612 (3.738)
$\mathbf{T}_j^{\text{US} \rightarrow \text{CN}} \times \text{PostTW}_t \times \text{CN Input Dep.}$		15.996*** (4.759)	21.023*** (5.622)	-0.545 (6.860)	31.395*** (6.879)	
$\mathbf{T}_j^{\text{US} \rightarrow \text{CN}} \times \text{PostTW}_t \times \text{CN Affiliate}$						4.836*** (1.864)
Observations	13,498	13,498	8,448	5,050	5,473	8,448
Pseudo R-squared	0.197	0.198	0.207	0.180	0.209	0.206

Notes: Manufacturing IMMEX firms operating as of 2017, over 2017–2024. Dependent variable is the number of new plants. Concurrent Trade Policy Controls include exposures to the U.S.–Mexico tariff escalation, China’s retaliatory tariffs, and the input channel measures $IMUS_j^{\text{CN} \rightarrow \text{US}}$, and $IMUS_j^{\text{US} \rightarrow \text{CN}}$, all constructed using firm-level data aggregated to the industry level and interacted with the post–trade war indicator. Columns (4)–(7) further control for Chinese Input Dependence $\mathbf{1} \left[\text{IMCN}_j^{\text{US} \rightarrow \text{CN}} \geq \text{Median} \right]$ interacted with the post–trade war indicator. CN Affiliate is a dummy variable that takes the value of one if the parent company headquarters are located in China, Hong Kong, or Singapore. All specifications include firm and border $\times$ year fixed effects. Robust standard errors are clustered at the three-digit industry level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

D.3 Robustness to USMCA-Affected Industries

The USMCA entered into force in July 2020 and introduced new rules that disproportionately affected a narrow set of manufacturing industries, most notably automotive and textile and apparel production. These provisions—including higher regional value content requirements and new labor value content thresholds in transportation equipment, as well as revised rules of origin in textiles and apparel—could, in principle, affect employment and production outcomes.

The baseline specification includes three-digit industry by year fixed effects, which absorb industry-wide shocks and time-varying trends common to narrowly defined sectors. As a result, broad adjustments associated with the implementation of the USMCA are already accounted for in the main estimates. Nevertheless, firms within the same three-digit industry may have been differentially affected by USMCA provisions due to heterogeneity in product composition, sourcing patterns, or integration into North American supply chains.

To assess whether such within-industry exposure to USMCA changes influences the results, I conduct a set of robustness exercises that re-estimate the employment and production effects after excluding industries most directly targeted by the agreement. Specifically, I sequentially exclude firms operating in textiles and apparel manufacturing (NAICS 313–315), transportation equipment manufacturing (NAICS 336), and the combined set of these industries. Across all specifications, production (Figure D.4a) and employment (Figure D.4b) effects remain positive and statistically significant, with no important differences, suggesting that the results reflect responses to the tariff shock rather than USMCA-related adjustments.

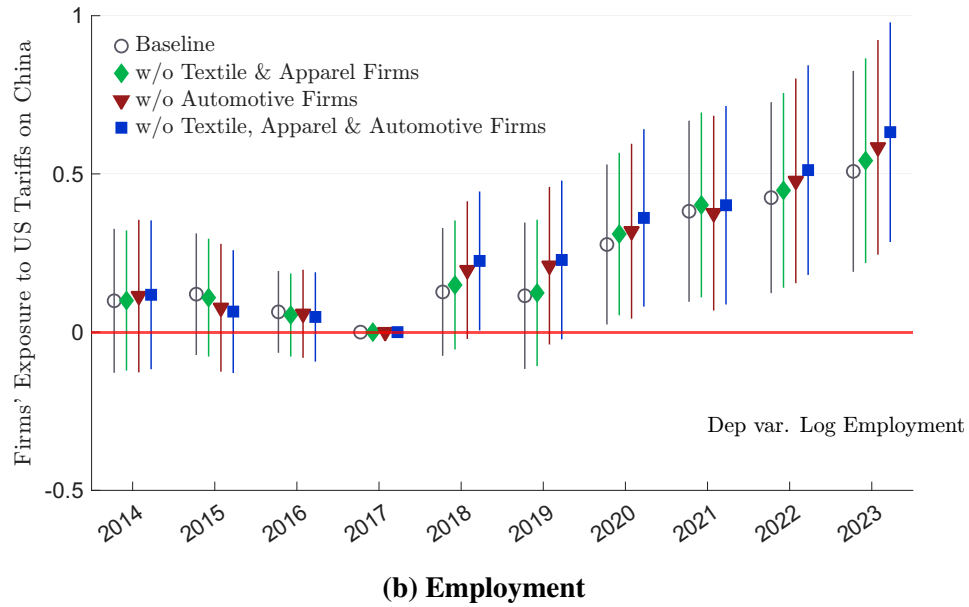
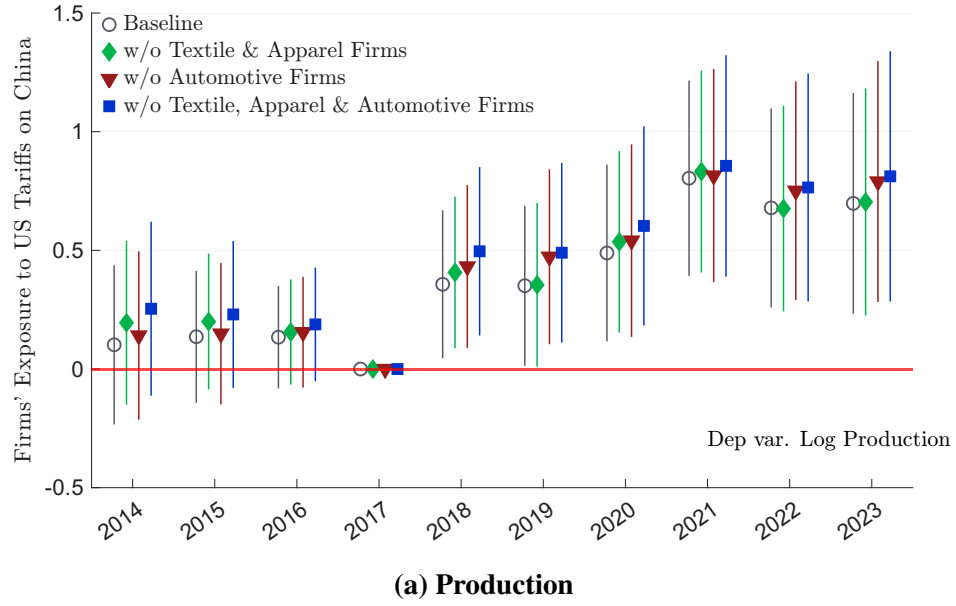


Figure D.4: Robustness to Excluding Firms Affected by USMCA Provisions

Notes: Estimation of Equation 3 using the treatment variable $T_{i,US}^{US \rightarrow CN}$ among 2017 U.S. exporters, as firms in textiles and apparel manufacturing (NAICS 313–315) and transportation equipment manufacturing (NAICS 336) are sequentially removed. The dependent variables are the natural logarithm of firms' value of production in (a) with $N=41,914/31,798$ and employment in (b) $N=42,171/31,970$. All regressions include firm fixed effects, three-digit industry-by-year fixed effects, and controls for concurrent trade policy changes. The y-axis displays yearly difference-in-differences coefficient estimates with 95% confidence intervals based on robust standard errors clustered at the firm level.